Generation of Chamber-Specific 3D Cardiac Organoids and an Integrated Electro-Mechanical Optical Analysis Platform

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August 2026

Approved by Keel Yong Lee, Ph. D.

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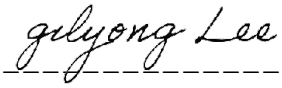
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Approved ----------

Keelyong Lee, Advisor

Abstract

Cardiovascular disease remains a leading cause of global mortality, yet the development of effective therapeutics is hindered by the limited predictive accuracy of conventional preclinical models. Human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) offer a physiologically relevant alternative, and three-dimensional (3D) culture further improves structural maturity over 2D monolayers. However, many generic 3D approaches still fail to recapitulate the chamber-specific electrophysiology of the native human heart, and a standardized platform for objectively quantifying the multi-modal functional output of engineered cardiac tissues is lacking. To address these gaps, we used a biphasic differentiation protocol with stage-specific retinoic acid (RA) modulation to generate chamber-specific 3D cardiac organoids — atrial (aCOs) and ventricular (vCOs). We developed **iCAMAnalyzer**, an integrated, Python-based platform that unifies intracellular Ca^{2+} transient kinetics, spatiotemporal activation mapping, and label-free mechanical contractility within a single graphical interface. The activation-mapping module was validated against synthetic ground-truth datasets, showing near-ideal recovery of conduction-velocity metrics ($R^2 > 0.99$), with propagation patterns cross-confirmed by independent frame-by-frame (ImageJ) analysis. Morphological and molecular assessment confirmed chamber-specific maturation: vCOs developed longer sarcomeres (~2.1 µm) with MLC2v and Cx43, whereas aCOs showed shorter sarcomeres (~1.7 µm) with MLC2a and Cx40. Using iCAMAnalyzer, we established quantitative "functional fingerprints" via Shape and Efficiency Indices: vCOs produced plateau-shaped Ca^{2+} transients (high Shape Index), whereas aCOs produced rapid, spike-shaped transients (low Shape Index, high Efficiency Index). Physiological fidelity was further supported by chamber-selective pharmacology — for example, carbachol elicited a selective negative chronotropic effect in aCOs. Finally, compatibility with a 3D-printed micro-pillar array demonstrated scalability toward high-throughput screening. In conclusion, this integrated optical platform for electro-

mechanical analysis provides a robust, objective, and physiologically relevant framework for precision cardiotoxicity screening and patient-specific disease modeling.

Keywords: in-vitro cardiac model[1], cardiovascular disease[2], tissue engineering[3], pluripotent stemcell[4], disease modeling[5], drug testing[6]

Contents

1. Introduction 10

2. Literature Review **12**

2.1 Historical Milestones in Cardiac Tissue Engineering 12

2.2 Paradigm Shift: From 2D Cultures to 3D Chamber-Specific Models 14

2.3 Structural and Functional Differences Between Atrial and Ventricular Tissues 17

2.4 Translating Chamber-Specific Principles into hiPSC-Derived Models 20

2.5 The Evolution of Functional Quantification Tools: The Necessity for a Digital Analytical Platform 27

3. Materials and Methods **29**

3.1 Human iPSC Culture and 2D (Monolayer) Cardiac Differentiation 29

3.2 Generation of Chamber-Specific 3D Cardiac Organoids 29

3.3 Fabrication of 3D-Printed Micro-pillar Arrays and Pacing Platform 31

3.4 Immunofluorescence Staining 33

3.5 Gene Expression Analysis by qRT-PCR 33

3.6 Flow Cytometry for Cardiomyocyte Purity 34

3.7 Optical Imaging for Ca^{2+} Transient Recording 34

3.8 iCAMAnalyzer: A Unified Python Platform for Ca^{2+} and Motion Analysis .. 35

3.9 Pharmacological Treatment 43
3.10 Cell Viability Assessment via Live/Dead Assay 43
3.11 Statistical Analysis 44

4. Results 47

4.1 An Integrated Optical Platform for Electro-Mechanical Analysis 47
4.2 Efficient Generation and Validation of Chamber-Specific Cardiomyocytes .. 47
4.3 Biphasic Differentiation and Morphological Evolution of 3D Cardiac Organoids 51
4.4 Morphological Characterization and Sarcomeric Maturation 53
4.5 Molecular Identity and Gap Junction Networking 55
4.6 Dynamic Calcium Transient Profiling and Functional Fingerprinting..... 57
4.7 Spatiotemporal Conduction and Electrical Pacing Fidelity 60
4.8 Label-Free Mechanical Contractility 63
4.9 Chamber-Specific Pharmacological Validation..... 66
4.10 Proof-of-Concept High-Throughput Multi-Organoid Platform..... 71

5. Discussion 74

5.1 Enhanced Structural and Functional Maturation in 3D Environments..... 74
5.2 Preservation of Chamber-Specific Identity via Timely Retinoic Acid Modulation 74

5.3 Parallel Quantification and Pharmacological Fingerprinting via iCAMAnalyzer 75

5.4 Limitations and Technical Constraints 75

5.5 Future Directions 76

6. Conclusion 78

7. References 79

8. Supplementary Figures 84

Korean Abstract 95

Acknowledgements 97

Lists of Figures

Figure 1. Major technological milestones in the evolution of cardiac tissue engineering

Figure 2. Structural hierarchy of the native heart and the comparative overview of in vitro cardiac platforms

Figure 3. Comparative structural and electrophysiological characteristics of ventricular-like and atrial-like cardiomyocytes

Figure 4. Developmental origins of chamber-specific cardiomyocytes and their derivation from hPSCs

Figure 5. Structural, molecular, and functional readouts for characterizing cardiac models

Figure 6. Design and fabrication of the 3D-printed micro-pillar array and field-pacing platform

Figure 7. Fluorescence-mode interface of iCAMAnalyzer for Ca^{2+} transient kinetics and spatiotemporal activation mapping

Figure 8. Automated calcium transient kinetic analysis and parameter extraction workflow

Figure 9. Automated mechanical motion and contractility analysis

Figure 10. Label-free quantification of contractile function by automated boundary tracking

Figure 11. Efficient generation and validation of chamber-specific cardiomyocytes

Figure 12. Biphasic differentiation protocol and morphological evolution of chamber-specific 3D cardiac organoids

Figure 13. Morphological characterization and sarcomeric maturation of chamber-specific 3D cardiac organoids

Figure 14. Molecular identity and gap junction networking of chamber-specific 3D cardiac organoids

Figure 15. Dynamic calcium transient profiling and functional index formulation for chamber-specific 3D cardiac organoids

Figure 16. Spatiotemporal conduction and pacing fidelity of chamber-specific 3D cardiac organoids

Figure 17. Label-free mechanical contractility of chamber-specific 3D cardiac organoids

Figure 18. Subtype-specific pharmacological sensitivity to Isoproterenol (ISO)

Figure 19. Pharmacological validation of atrial-specific functional response to Carbachol

Figure 20. Proof-of-concept high-throughput, multi-organoid analysis on the micro-pillar platform

Lists of Supplementary Figures

Supplementary Figure S1. Induction of chamber-specific cardiomyocytes from hiPSCs via dual Wnt signaling modulation

Supplementary Figure S2. Optimization of initial cell seeding density for stable cardiac organoid formation

Supplementary Figure S3. Automated morphological quantification via the custom-developed Organoid Diameter Analyzer

Supplementary Figure S4. Differentiation timelines and aggregation strategies for chamber-specific cardiac spheroids and organoids

Supplementary Figure S5. Characterization of cell viability in chamber-specific cardiac organoids

Supplementary Figure S6. Calcium transient profiling and functional indices of chamber-specific cardiac spheroids

Supplementary Figure S7. Quantitative in silico validation of the conduction-velocity and activation-mapping module

Supplementary Figure S8. Application and cross-method validation of the activation-mapping module on a real cardiac organoid

Lists of Tables

Table 1. Comprehensive characterization of chamber-specific cardiac tissues: from molecular markers to functional phenotypes

Table 2. Integrated functional and technical comparison of iCAMAnalyzer with existing cardiac optical analysis tools

Table 3. List of antibodies and fluorescent stains used for immunofluorescence

Table 4. Primer sequences used for qPCR

Table 5. Ca^{2+} analysis parameters extracted by iCAMAnalyzer

I. Introduction

Cardiovascular diseases (CVDs) remain the leading cause of mortality worldwide, accounting for approximately one in three global deaths in 2023[1]. Yet the translation of laboratory findings into clinical therapies is highly inefficient: roughly 90% of drug candidates fail during clinical development, with cardiotoxicity ranking among the leading causes of attrition and post-market withdrawal[2,3]. This systemic failure stems fundamentally from the limitations of conventional preclinical models. Traditionally, in vivo animal models have been utilized as the standard for evaluating drug safety; however, they exhibit significant interspecies mismatches. For instance, resting heart rates of rodents (~300–700 BPM in rats and mice) differ vastly from those of humans (~60–100 BPM)[4,5], alongside fundamental discrepancies in ion channel expression profiles. Consequently, extrapolating drug responses from animal models to human physiology yields poor predictive accuracy.

To mitigate these interspecies issues, in vitro approaches have become essential. For preclinical cardiotoxicity evaluation, single-ion-channel assays (e.g., the hERG block assay in heterologous expression systems)[6,7] and 2D hiPSC-CM monolayer platforms have been widely employed[7,8]. While the integration of human induced pluripotent stem cell-derived cardiomyocytes (hiPSC-CMs) into 2D platforms has helped bridge the interspecies genetic gap, these models fundamentally lack 3D mechanical complexity and exhibit an immature structural phenotype[9,10]. More critically, 2D hiPSC-CM models rely on isolated electrophysiological readouts. Relying solely on isolated single-channel data neglects multi-channel interactions, frequently resulting in false-positive or false-negative predictions that disrupt the drug development pipeline[11,12].

Consequently, three-dimensional (3D) engineering of hiPSC-CMs has emerged as an increasingly viable, human-relevant alternative to overcome both the interspecies mismatches of animal models and the physical limitations of 2D cultures. This transition aligns with recent regulatory paradigm shifts, such as the FDA Modernization Act 2.0

and the Comprehensive In Vitro Proarrhythmia Assay (CiPA) initiative. However, current hiPSC-CM platforms possess two critical limitations.

First, most generated models remain generic, structurally immature, and heavily ventricular-biased[9,13,14]. The working myocardium of the atria and ventricles performs the bulk of cardiac contractile work, and the two chambers differ markedly in physiological role, electrophysiology, and disease susceptibility[15,16]. This necessitates the development of independent, standardized chamber-specific models.

Second, a standardized analytical framework capable of evaluating integrated excitation-contraction (E-C) coupling remains absent. Existing approaches analyze electrophysiological signals and mechanical contraction in isolation[17,18], thereby failing to capture integrated arrhythmia mechanisms[19].

Recognizing these critical gaps, this study aims to (i) engineer chamber-specific 3D cardiac organoids and (ii) establish a dual-module analytical platform that resolves Ca^{2+} kinetics, electrical conduction, and mechanical contractility through two complementary, dedicated programs. The specific objectives of this study are fourfold:

1. **Generation of Chamber-Specific Models:** To generate atrial and ventricular 3D organoids utilizing a biphasic protocol combined with precise Wnt and retinoic acid (RA) modulation.
2. **Multimodal Analysis Platform:** To develop iCAMAnalyzer, a Python-based integrated analytical platform that unifies intracellular Ca^{2+} transient kinetics, spatiotemporal activation mapping, and label-free mechanical contractility analysis within a single graphical user interface.
3. **Physiological Validation:** To validate the functional fidelity and chamber-specificity of the models using targeted pharmacological agents (Carbachol and Isoproterenol).
4. **High-Throughput Scalability:** To demonstrate proof-of-concept multi-organoid acquisition on custom 3D-printed micro-pillar arrays, laying the foundation for scalable screening.

II. Literature Review

2.1. Historical Milestones in Cardiac Tissue Engineering

Over the past decades, technological milestones have driven the evolution of in vitro cardiac systems toward recapitulating the native myocardium **(Fig. 1)**. Since the late 1990s the field has progressed from scaffold-based constructs to three-dimensional (3D) engineered heart tissues (EHTs) and microphysiological systems (MPS). Foundational work in scaffold-based myocardial grafting[20] and the introduction of hiPSC-derived cardiomyocytes[21] established the cellular and structural building blocks, while later platforms refined electrical and mechanical conditioning to drive functional maturation[22,23]. At the macroscopic level, rotational molding and 3D bioprinting produced ventricular models that replicated chamber curvature and wall architecture—for example, MacQueen et al.'s pull-spun scale model of the human left ventricle, which recapitulated macroscopic pumping dynamics[24]—culminating recently in the first transplantation of hiPSC-derived engineered heart muscle in non-human primates and in patients with heart failure[25]. While these macroscopic models demonstrate global biomechanical function, they require large cell numbers and remain incompatible with high-throughput pharmacological screening.

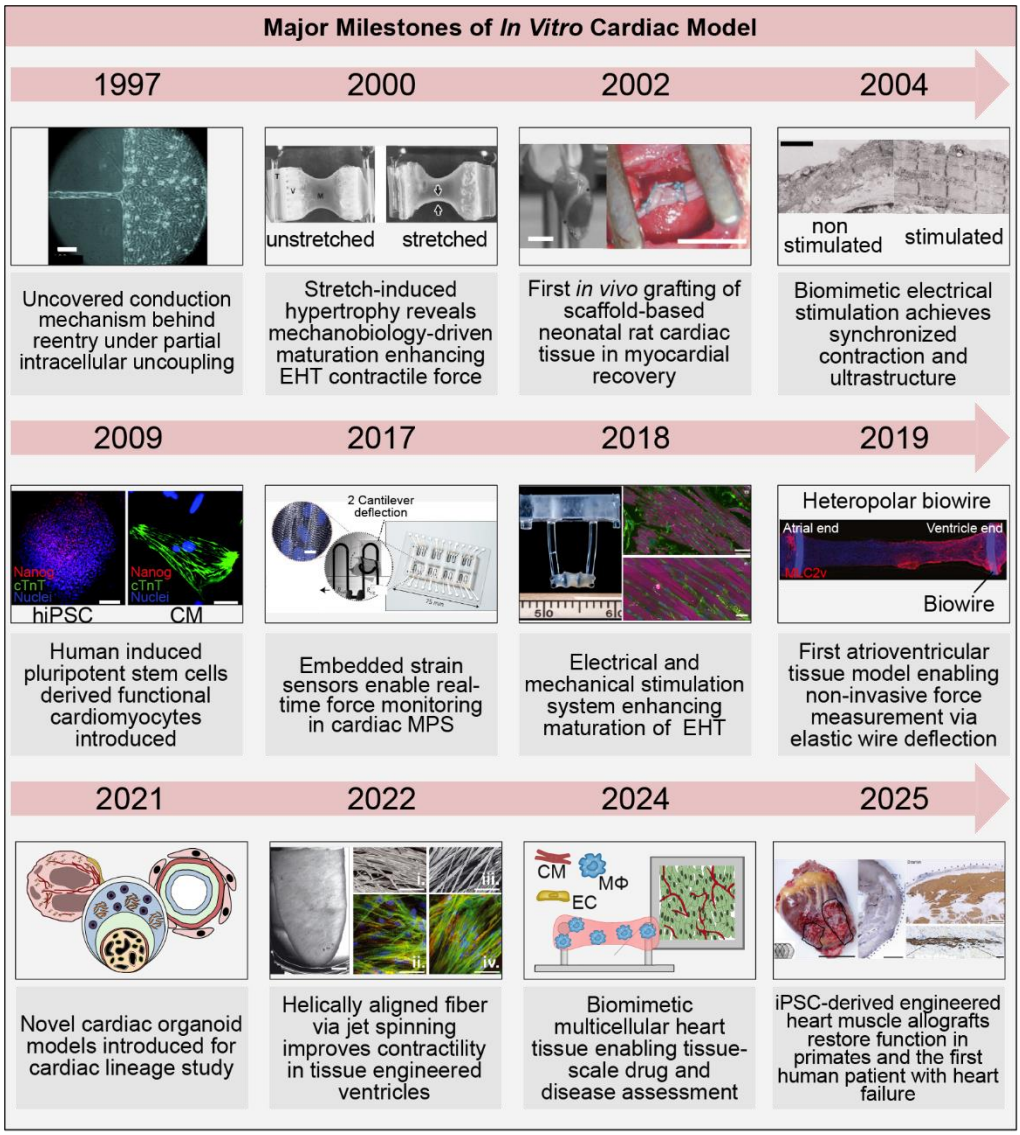


Fig. 1 Major technological milestones in the evolution of cardiac tissue engineering. A historical timeline (1997–2025) of key advances in cardiac tissue engineering. Scale bars are shown for each representative image: (scale bar = 200 μm) [26] (1991), (scale bar = 100 μm) [27](1997), [28](2000), (scale bar = 1 cm) [29](2002), (scale bar = 2 μm) [30](2004), [scale bars = 100 μm (left), 20 μm (right)] [31] (2009), [23](2017), [scale bars = 100 μm (right-top), 20 μm (right-bottom)] [22](2018), (scale bar = 0.5 mm) [32](2019), (2021), [scale bars = 2 mm (left), 50 μm (right)] [33](2022), (2024), [scale bars = 2 mm (left), 1 cm (middle), 2 mm (right-top), 200 μm (right-bottom)] [25](2025).

2.2 Paradigm Shift: From 2D Cultures to 3D Chamber-Specific Models

To overcome the inherent biomechanical limitations of early two-dimensional (2D) planar cultures and macroscopic models, recent bioengineering trends have fundamentally shifted toward micro-scale 3D platforms that better reflect the native structural hierarchy (**Fig. 2**). Among these, three architectural strategies dominate: rod- or wire-shaped EHTs, self-organizing organoids, and organ-on-chip MPS. Each strategy offers a distinct balance of structural control, throughput, and physiological complexity, and the choice of platform directly shapes the kinds of biological questions that can be addressed.

A representative wire-based platform is **Biowire II**, developed by Zhao et al., which generates spatially distinct atrial- and ventricular-specific tissues suspended between two flexible poly(octamethylene maleate (anhydride) citrate) (POMaC) wires within a polystyrene microwell array. By combining directed differentiation with long-term electrical conditioning, Biowire II achieves robust chamber-specific specification and enables simultaneous quantification of contractile force and Ca^{2+} transients, making it highly effective for chamber-selective drug response testing[32].

Leveraging the principles of developmental biology, the field has expanded to include **multi-chamber cardioids** that harness the intrinsic self-organizing capabilities of pluripotent stem cells. Schmidt et al. recently developed advanced multi-chamber cardioids by temporally modulating Wnt, BMP, FGF, and retinoic acid signaling to differentiate the first heart field, anterior second heart field, and posterior second heart field separately, then fusing them into integrated atrial–ventricular structures that recapitulate directional electrical conduction[34]. These self-organizing platforms provide high-resolution, developmentally relevant microenvironments and offer unprecedented opportunities for modeling chamber-specific disease phenotypes.

Between these two ends — engineered tissues with strict geometric control and self-organizing cardioids with developmental fidelity — lies **3D cardiac organoid** approaches that combine biphasic differentiation with physical anchoring (e.g., micro-

pillar arrays). This intermediate approach was selected for the present study because it preserves the self-organization advantage of organoid culture, scales to 96-well or higher throughput formats, and requires far fewer cells per tissue than macroscopic EHTs — properties essential for chamber-specific pharmacological screening at the scale required by modern regulatory frameworks.

Indeed, this transition aligns with recent regulatory paradigm shifts. The **Comprehensive In Vitro Proarrhythmia Assay (CiPA)** initiative, launched by the FDA in 2013, explicitly calls for multi-ion-channel evaluation, in silico reconstruction of ventricular electrophysiology, and validation in hiPSC-derived cardiomyocytes — moving beyond the hERG-only paradigm of ICH S7B and E14[12,35]. Together with the FDA Modernization Act 2.0 (2022), which permits non-animal alternatives in regulatory submissions, these shifts have created direct demand for 3D, chamber-specific hiPSC-CM platforms paired with integrated functional readouts.

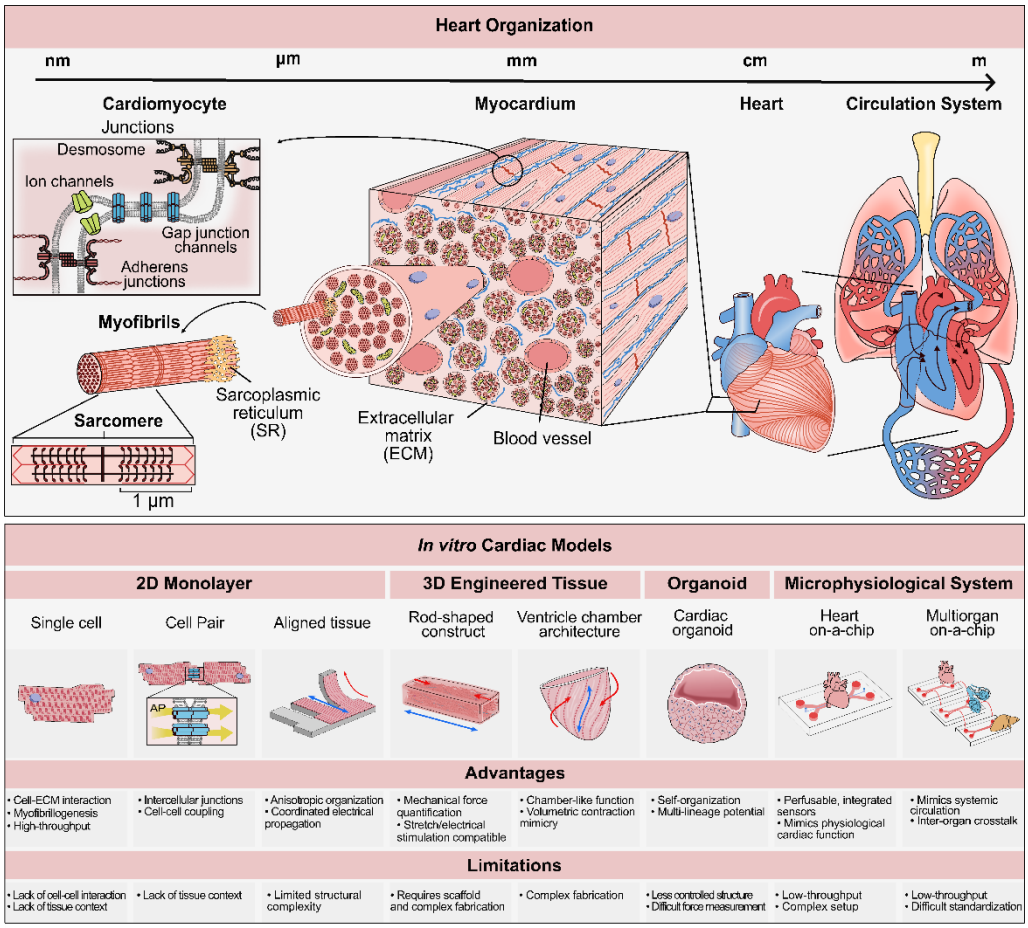
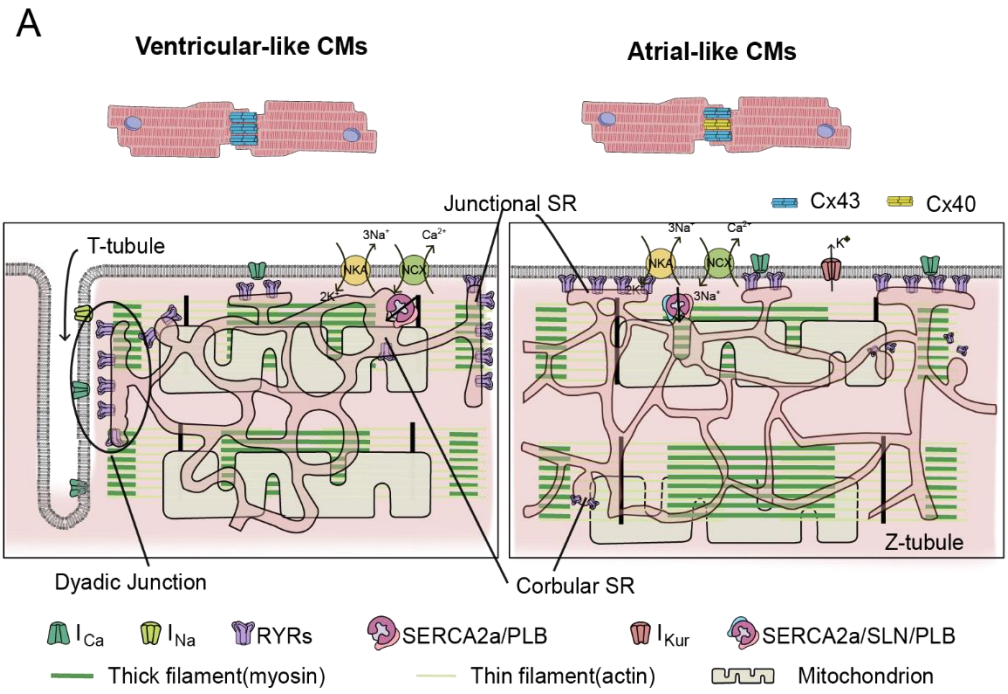


Fig. 2 Structural hierarchy of the native heart and the comparative overview of *in vitro* cardiac platforms. (Top) Schematic representation of the human cardiovascular system's hierarchical organization. (Bottom) The paradigm shift from conventional 2D monolayers to 3D engineered heart tissues and cardiac organoids, outlining their respective bioengineering advantages and limitations. Red arrows indicate contraction direction; blue arrows, relaxationdirection. SR sarcoplasmic reticulum, ECM extracellular matrix, AP action potential, EHT engineered heart tissue, hiPSC human induced pluripotent stem cells, CM cardiomyocyte, MPS microphysiological system, МΦ macrophage, EC endothelial cell.

2.3 Structural and Functional Differences Between Atrial and Ventricular Tissues

The atrium and ventricle are functionally divergent by design, and this study therefore targets the two chambers specifically rather than treating cardiac tissue as a homogeneous entity. The atrium is a thin-walled, low-pressure primer pump with a fast intrinsic rate and a short, triangular action potential (AP); it is the principal site of atrial fibrillation (AF), the most prevalent arrhythmia. The ventricle is a thick-walled, high-pressure pump with a markedly longer AP dominated by a plateau phase, within which pathological QT prolongation and Torsades de Pointes (TdP) arise. Because AF and TdP originate from distinct chamber substrates, dissecting either pathology—and verifying that an anti-AF drug modulates atrial channels without provoking ventricular QT prolongation via hERG/IKr block—requires a platform that assays the two chambers independently and in parallel. Chamber-specific modeling is thus a prerequisite for safe preclinical drug evaluation (Fig. 3, Table 1).

These organ-level differences are rooted in distinct cellular architecture, ion currents, and gap junctions. Structurally, ventricular cardiomyocytes are large and rod-shaped with a dense transverse-tubular (T-tubule) network forming dyadic junctions with the sarcoplasmic reticulum (SR) for synchronous, spatially uniform Ca^{2+} release; atrial cardiomyocytes are smaller, lack an organized T-tubule system, and rely on junctional/corbular SR, so Ca^{2+} propagates more slowly from the periphery inward—yielding faster but weaker transients. Electrophysiologically, the ventricular AP shows a prolonged plateau (Phase 2) sustained by inward I_Ca,L and a high density of the inward-rectifier I_{K1} (KCNJ2), giving a long APD90 and slower intrinsic rate; the atrial AP is shorter and triangular, governed by atrial-selective currents largely absent from ventricle—the ultra-rapid delayed rectifier I_{Kur} (Kv1.5/KCNA5) and the acetylcholine-activated $I_{K,ACh}$ (Kir3.1/3.4, KCNJ3/5). Molecularly, ventricular cells express MLC2v (MYL2) and Cx43 (GJA1), whereas atrial cells retain MLC2a (MYL7) and co-express Cx40 (GJA5) with Cx43, supporting faster atrial conduction; the transcription factors NR2F2 (atrial) and IRX4 (ventricular) mark lineage identity (Fig. 3; Table 1).


A
Ventricular-like CMs
Atrial-like CMs
Junctional SR
Cx43
Cx40
T-tubule
NKA
NCX
Dyadic Junction
Corbular SR
Z-tubule
I Ca
I Na
RYRs
SERCA2a/PLB
I Kur
SERCA2a/SLN/PLB
Thick filament(myosin)
Thin filament(actin)
Mitochondrion

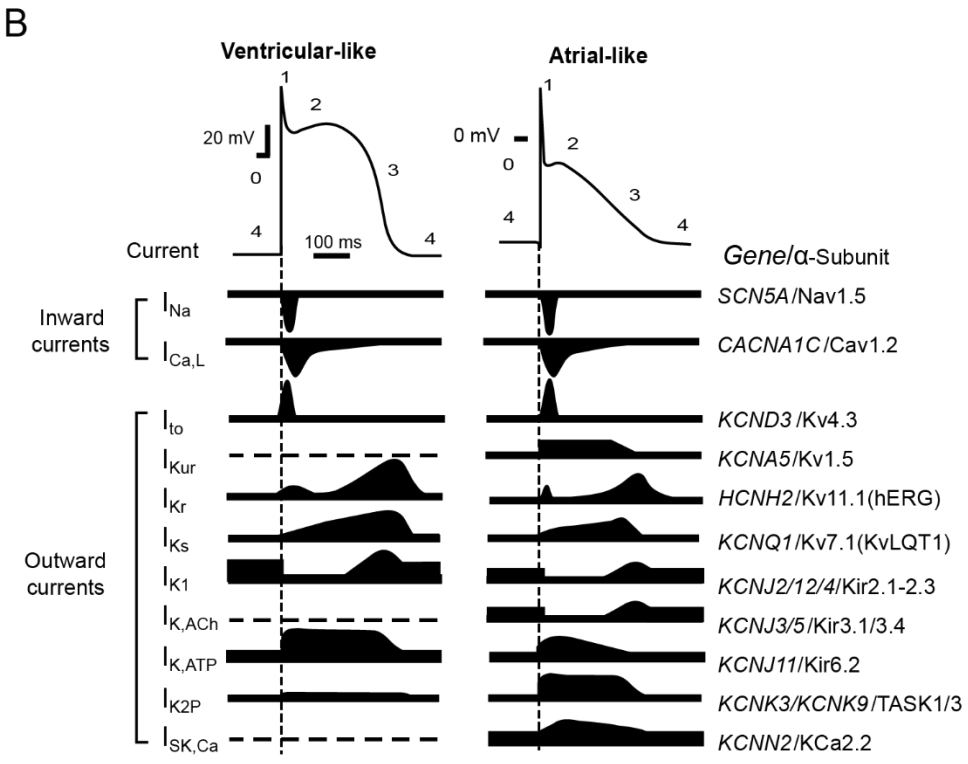

B
Ventricular-like
Atrial-like
20 mV
0 mV
100 ms
Current
Gene/α-Subunit
Inward currents
Outward currents
I Na
I Ca,L
I to
I Kur
I Kr
I Ks
I K1
I K,ACh
I K,ATP
I K2P
I SK,Ca
SCN5A/Nav1.5
CACNA1C/Cav1.2
KCND3/Kv4.3
KCNA5/Kv1.5
HCNH2/Kv11.1(hERG)
KCNQ1/Kv7.1(KvLQT1)
KCNJ2/12/4/Kir2.1-2.3
KCNJ3/5/Kir3.1/3.4
KCNJ11/Kir6.2
KCNK3/KCNK9/TASK1/3
KCNN2/KCa2.2

Fig. 3 Comparative structural and electrophysiological characteristics of ventricular-like and atrial-like cardiomyocytes. (A) Schematic representation of structural organization and calcium-handling machinery. Ventricular-like cardiomyocytes (CMs) are characterized by a well-developed T-tubule system and dyadic junctions, facilitating synchronized calcium release. In contrast, atrial-like CMs lack distinct T-tubules, relying on junctional and corbular sarcoplasmic reticulum (SR) for excitation-contraction coupling. Distinct gap junction protein expression is also highlighted, with Connexin 43 (Cx43) predominant in the ventricles and Connexin 40 (Cx40) in the atria. (B) Profiles of action potentials (APs) and underlying ionic currents. Ventricular APs exhibit a prominent plateau phase (Phase 2), primarily driven by sustained inward $I_{Ca,L}$ and higher I_{K1} density. Atrial APs exhibit a more triangular morphology with a shorter duration, characterized by atrial-specific currents including I_{Kur}, $I_{K,ACh}$, and $I_{SK,Ca}$. Corresponding genes and alpha-subunits for each inward and outward current are indicated on the right.

2.4 Translating Chamber-Specific Principles into hiPSC-Derived Models

With the paradigm-shifting advent of human induced pluripotent stem cell (hiPSC) technology, the developmental signaling pathways discovered *in vivo* have been leveraged to generate subtype-specific human cardiomyocytes *in vitro*. Under ideal directed differentiation protocols, hiPSC-derived atrial cardiomyocytes (hiPSC-aCMs) recapitulate their native counterparts by expressing NR2F2, exhibiting rapid Ca^{2+} transient decay, and presenting a lower baseline contractile force. Conversely, hiPSC-derived ventricular cardiomyocytes (hiPSC-vCMs) express IRX4 and MYL2, display a prolonged APD_{90}, and generate substantially higher inotropic outputs (Table 1).

However, translating these native principles into stable *in vitro* screening platforms presents two major bioengineering hurdles:

1. Default Ventricular Bias and Progenitor Heterogeneity: Default, non-directed hiPSC differentiation protocols predominantly yield ventricular-like cells (90% $cTNT^+$ ventricular fate). This bias stems from the intrinsic signaling trajectories of early cardiac progenitor populations during cardiogenesis. The mammalian heart develops from two distinct populations of mesodermal progenitors: the First Heart Field (FHF) and the Second Heart Field (SHF), the latter of which is further sub-divided into the anterior SHF (aSHF) and posterior SHF (pSHF)[13,14].

- The First Heart Field (FHF) & Anterior Second Heart Field (aSHF): The FHF primarily gives rise to the primitive heart tube, contributing significantly to the left ventricle and portions of the atria. The aSHF progenitors, regulated by transcription factors such as ISL1 and TBX1, migrate into the anterior pole of the heart tube to form the right ventricle and the outflow tract. Crucially, both FHF and aSHF lineages differentiate into working ventricular myocardium in an environment naturally shielded from retinoic acid (RA) signaling.
- The Posterior Second Heart Field (pSHF): In contrast, the pSHF constitutes a molecularly distinct progenitor pool located at the caudal end of the splanchnic mesoderm. Driven by the targeted integration of HOX gene expressions and

specific transcriptional regulators, pSHF progenitors actively migrate into the posterior venous pole of the developing heart to form the definitive atrial working myocardium and the inflow tract.

2. Retinoic Acid Signaling and the Window of Atrial Specification: Because chamber identity is fundamentally determined by these spatially segregated progenitor pools, replication of this lineage commitment *in vitro* requires precise manipulation of the retinoic acid (RA) morphogen gradient[36,37]:

- **RA Concentration and Presence:** While FHF and aSHF-derived ventricular lineages develop in the complete absence of RA, exogenous RA signaling is absolutely mandatory to specify the pSHF lineage toward an atrial fate. Exposure to RA actively activates the expression of the orphan nuclear receptor NR2F2, which suppresses the ventricular genetic program (such as IRX4 and HAND1) within the expanding pSHF population. Uncontrolled or excessive RA exposure, however, forces the entire progenitor pool toward a pSHF fate, causing hyperplastic atrial formation at the expense of ventricular structures.
- **Timing of Exposure:** The developmental window during which cardiac progenitors remain responsive to RA is highly restricted. In hiPSC models, activation of the RA pathway must occur strictly during the cardiac mesoderm specification stage (typically between Days 3 and 5 of differentiation). Administering RA after Day 5 fails to induce the pSHF identity, as the cells have already committed irreversibly to the default, FHF/aSHF-like ventricular lineage.

3. Fetal-Like Structural Immaturity: Beyond lineage specification, standard monolayer-derived hiPSC-CMs frequently present a fetal-like, structurally immature phenotype regardless of the directed protocols used. A prominent hallmark of this immaturity is the abnormal co-expression of MLC2a and MLC2v within the same single cell, alongside a complete lack of organized T-tubules[9,38].

These baseline discrepancies and developmental reference ranges define the precise operating window against which the engineered models must be benchmarked. To

bridge this developmental gap, utilizing three-dimensional (3D) physical constraints and tissue-engineering scaffolding remains absolutely crucial. Driving functional maturation requires transitioning from flat 2D surfaces to 3D microphysiological environments that can enforce structural alignment, mimic native mechanical load, and promote chamber-specific adult-like phenotypes derived from their respective FHF/aSHF or pSHF origins.

Comprehensive evaluation of engineered cardiac tissues draws on three complementary categories of readout (Fig. 5) Structural readouts assess macro-scale muscle alignment and chamber formation together with micro-scale sarcomeric organization, extracellular-matrix architecture, gap-junction distribution, and T-tubule development. Molecular readouts, increasingly delivered by spatial and single-cell transcriptomics, resolve chamber- and maturation-associated gene programs—including diagnostic isoform switches such as MYH6→MYH7, TNNI1→TNNI3, and MYL7→MYL2—and can infer intercellular communication. Functional readouts capture the physiological output of the tissue: electrophysiology (action-potential morphology, APD, and arrhythmia/re-entry mapping), mechanics (excitation–contraction coupling, calcium-transient kinetics, and contractility), and metabolism (the fetal-to-adult shift from glycolytic toward oxidative energetics). Among these, functional readouts are the most direct reporters of chamber-specific physiology and drug response, yet remain the most technically demanding to quantify in three dimensions—motivating the integrated optical analysis platform developed in this study.

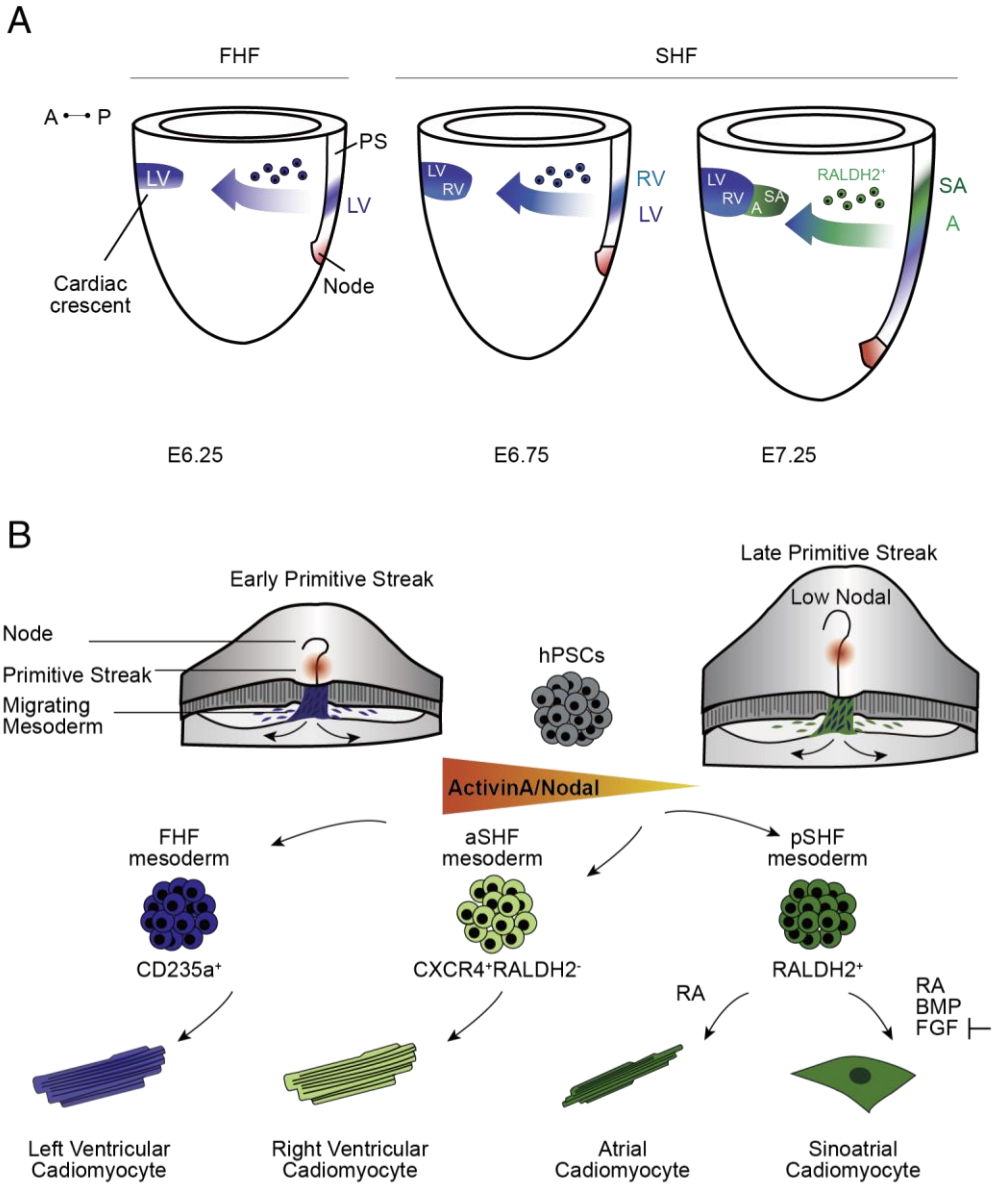

A
FHF
SHF
A
P
PS
LV
LV
Cardiac
crescent
Node
LV
RV
RV
LV
LV
RV
A
SA
RALDH2+
SA
A
E6.25
E6.75
E7.25
B
Early Primitive Streak
Late Primitive Streak
Low Nodal
Node
Primitive Streak
Migrating
Mesoderm
hPSCs
ActivinA/Nodal
FHF
mesoderm
aSHF
mesoderm
pSHF
mesoderm
CD235a+
CXCR4+RALDH2-
RALDH2+
RA
RA
BMP
FGF
Left Ventricular
Cadiomyocyte
Right Ventricular
Cadiomyocyte
Atrial
Cadiomyocyte
Sinoatrial
Cadiomyocyte

Fig. 4 Developmental origins of chamber-specific cardiomyocytes and their derivation from hPSCs. (A) In vivo mouse heart-field development. The first heart field (FHF) forms the cardiac crescent (E6.25), while the second heart field (SHF) contributes progressively (E6.75–E7.25), with RALDH2^{+} posterior progenitors giving rise to atrial/sinoatrial regions. (B) In vitro recapitulation. Under a graded Activin A/Nodal signal, hPSC-derived mesoderm is patterned into, anterior, and posterior SHF; retinoic acid (RA) directs posterior progenitors toward atrial cardiomyocytes, whereas combined RA/BMP/FGF signaling specifies sinoatrial fate.

Table 1. Comprehensive characterization of chamber-specific cardiac tissues: from molecular markers to functional phenotypes.

Category	Parameter	Atrial(Atrial-like)	Ventricular (Ventricular-like)	Ref.
Molecular Markers	Transcription Factors	*NR2F2*, *PITX2*	*IRX4*, *HAND1*	[39]
	Structural Proteins	*MYL7* (MLC-2a), *NPPA* (ANP)	*MYL2* (MLC-2v), *MYH7*	[32,34]
	Ion Channels	*KCNA5* (I_{Kur}), *KCNJ3/5* ($I_{K,ACh}$)	*KCNJ2* (I_{K1}), *SCN5A* (I_{Na})	[39] [32]
Gap Junctions	Connexins	Cx40 (*GJA5*), Cx43	Cx43 (*GJA1*)	[40]
Electrophysiology	APD_{90} (Native)	150~200 ms	250 ~ 400 ms	[39]
	APD_{90} (3D COs)	100 ~ 200 ms	300 ~ 450 ms	[32]
	Beating Rate	60 ~ 90 bpm (Faster)	20 ~ 50 bpm (Slower)	[34]
Calcium Kinetics	RT_{50} (Native)	100 ~ 150 ms	200 ~ 300 ms	[40]
	RT_{50} (3D COs)	150 ~ 250 ms	300 ~ 450 ms	[32]
Biomechanics	Contractile Force	~ 0.5mN/mm^2	1.0 ~ 4.0 mN/mm^2	[32]

Native; Native Human, 3D COs; 3D hiPSC-derived cardiac organoids

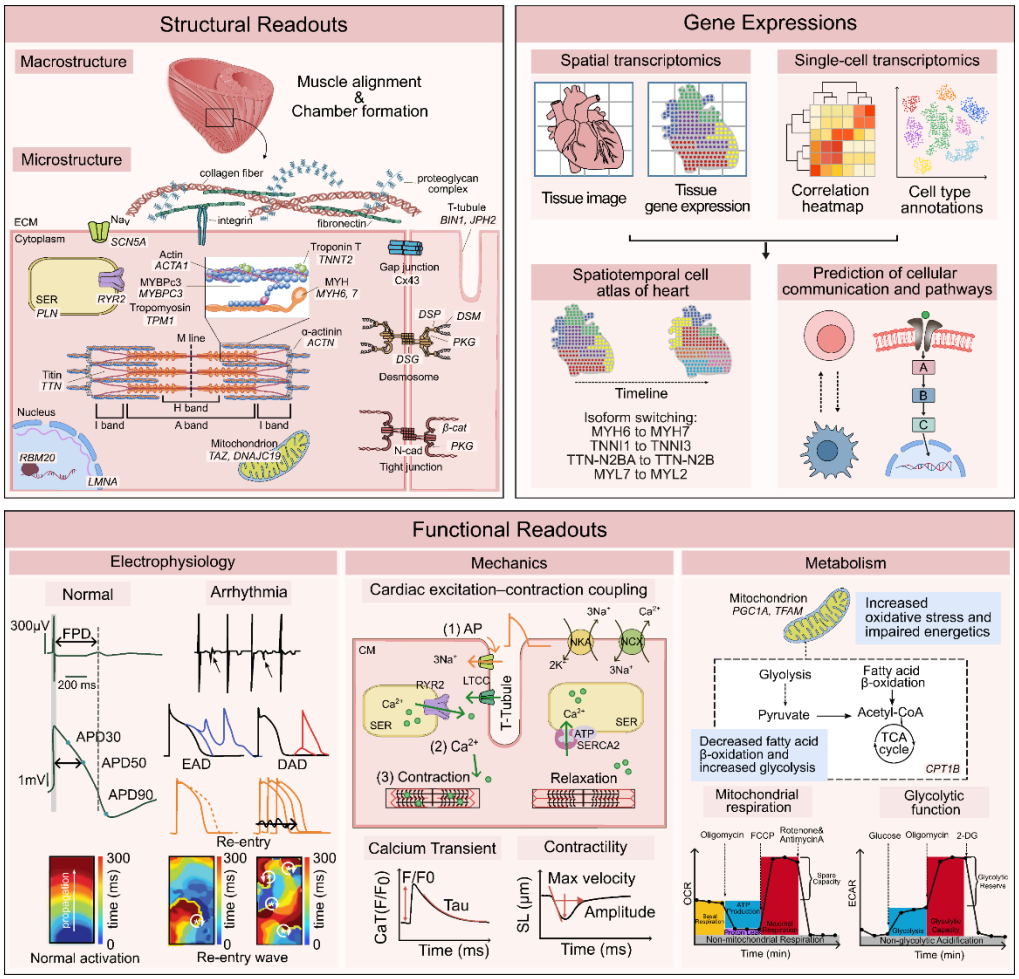


Fig. 5 Structural, molecular, and functional readouts for characterizing cardiac models. (Structural) Macro- and micro-structural assessment of muscle alignment, chamber formation, sarcomeric and ECM organization, gap junctions, and T-tubules, with associated genes. (Gene expression) Spatial and single-cell transcriptomics yielding spatiotemporal cell atlases, isoform-switching signatures (e.g., MYH6→MYH7, TNNI1→TNNI3, MYL7→MYL2), and inferred cell–cell communication. (Functional) Electrophysiology (action-potential/APD metrics, arrhythmia and re-entry mapping), mechanics (excitation–contraction coupling, calcium transients, contractility), and metabolism (oxidative versus glycolytic flux by OCR/ECAR).

2.5 The Evolution of Functional Quantification Tools: The Necessity for a Digital Analytical Platform

Chamber-specific 3D organoids yield complex constructs whose physiological traits—Ca^{2+}-transient waveforms and mechanical contraction patterns—must be translated into reproducible, quantitative metrics (e.g., Shape Index, Fractional Area Change) rather than qualitative observations, requiring software that processes high-speed optical data and resolves the kinetic differences between atrial and ventricular phenotypes. Traditionally, this quantification has relied on a fragmented tool ecosystem: MATLAB-based packages (e.g., CalTrack, COSMAS) for calcium/optical electrophysiology, and the Java/ImageJ MUSCLEMOTION macro or custom Python scripts for contractility. Operating unlinked programs across MATLAB, Java, and Python introduces data-interoperability hurdles, manual reformatting of large high-speed video datasets, and commercial-license costs that limit high-throughput deployment. To address this, the present study introduces iCAMAnalyzer, an all-inclusive Python platform. Rather than extracting coupled excitation–contraction kinetics from a single dual-channel recording—which demands optical splitters and complex configurations—iCAMAnalyzer integrates dedicated modules for intracellular Ca^{2+} kinetics and label-free macro-scale motion tracking within a single graphical interface, processing independent high-speed datasets under a standardized algorithmic routine. This minimizes software friction, removes commercial-license dependencies, and provides a streamlined toolchain for establishing quantitative "functional fingerprints" of engineered cardiac tissues (Table 2).

Table 2. Integrated Functional and Technical Comparison of iCAMAnalyzer with Existing Cardiac Optical Analysis Tools

Software Tool	Ca^{2+} Kinetics	CV / Activation	Spontaneous (FFT)	Motion (Brightfield)	Arrhythmia Panel	Core Capability / Key Limitation
iCAMAnalyzer *(present study)* Python · Open-source	✓	✓	✓	✓	Δ	Only tool unifying Ca^{2+} kinetics and brightfield motion in a single dual-module GUI. Organoid-scale arrhythmia panel (EAD/DAD, alternans) implemented; validation ongoing (Δ).
COSMAS MATLAB · Scripting	✓	✓	—	—	—	High-fidelity activation/CV and AP–CaT analysis for paced optical mapping; no GUI, motion, or spontaneous-beat detection[41].
ElectroMap MATLAB · Open-source	✓	✓	—	—	✓	High-throughput optical-mapping suite with alternans quantification; no motion module; optimized for tissue- (*ex vivo*), not organoid-scale[42].
CalTrack MATLAB	✓	—	✓	—	✓	Automated Ca^{2+} kinetics with waveform-abnormality classification; no spatial CV/activation mapping or motion tracking[43].
mps-motion Python · Open-source	—	—	✓	✓	—	Optical-flow brightfield displacement/velocity tracking; no calcium or CV/activation mapping[42].
MUSCLEMOTION ImageJ macro · Open-source	—	—	✓	✓	—	Widely used brightfield contractility plugin; no calcium kinetics or CV mapping; requires ImageJ runtime[17].

✓, capability supported; —, not supported; Δ, implemented, pending experimental validation. CV, conduction velocity; FFT, fast Fourier transform-based spontaneous beating detection.

III. Materials and Methods

3.1. Human iPSC Culture and 2D (Monolayer) Cardiac Differentiation

Human induced pluripotent stem cells (hiPSCs; iPSCdirect™ SCTi003-A, STEMCELL Technologies) were maintained in Essential 8 medium (Gibco) on 6-well plates pre-coated with a mixture of 200 µL Geltrex (Gibco) and 200 µL human plasma fibronectin (1 mg/mL; Gibco) in 40 mL PBS (Gibco). At 70–80% confluency (~3 days), cells were passaged at 1:6 using Versene (Gibco). During passaging and thawing, the medium was supplemented with the ROCK inhibitor Y-27632 (STEMCELL Technologies; 5 µM). For 2D monolayer cardiac differentiation, a chemically defined Wnt-modulation protocol was applied **(Fig. 11A; Supple Fig. S1).** At 80–90% confluency (Day 0), hiPSCs were treated with 6 µM CHIR99021 in RPMI 1640/B27-minus-insulin for 48 h, rested in the same medium without CHIR99021 for 24 h, and then treated with 5 µM IWR-1-endo for 48 h. For chamber specification, atrial-like cells were induced with 1 µM retinoic acid (RA; Sigma-Aldrich, R2625) during (Day 3–5), whereas ventricular-like cells were generated by omitting RA. Spontaneous beating appeared by Day 6–8. Cardiomyocytes were then enriched by lactate metabolic selection (glucose-free RPMI + 5 mM sodium lactate, 2–4 days) between Day 15 and 20, and maintained in Advanced MEM (Gibco, 12492021) supplemented with 1% penicillin–streptomycin (Gibco, 15140-122) and 1% GlutaMAX (Gibco, 35050061) for long-term culture and maturation. Differentiation efficiency and chamber identity of the 2D cardiomyocytes were confirmed by flow cytometry (Section 3.6), qPCR (Section 3.5), and Ca^{2+}-transient imaging (Section 3.7) **(Fig. 11B-D).**

3.2. Generation of Chamber-Specific 3D Cardiac Organoids

To overcome the structural and physiological limitations of 2D monolayer cultures, chamber-specific 3D **cardiac** organoids (COs) were generated using two aggregation

platforms **(Supple Fig. S4A, B):** an ultra-low-attachment (ULA) plate method and an orbital-shaker/Matrigel method. The transition to 3D partially recapitulates the structural hierarchy and functional compartmentalization of the native myocardium.

3.2.1 Cardiac Differentiation and Chamber-Specific Specification

All approaches used the same chemically defined dual Wnt-modulation protocol **(Fig. 12B).** Mesendoderm was induced from Day 0 with 6 μM CHIR99021 in RPMI 1640/B27-minus-insulin, followed by 5 μM IWR-1-endo from Day 2 to Day 4. For chamber specification, 1 μM retinoic acid (RA) was co-administered during subtype specification to drive atrial-like identity — with a method-specific window **(Fig. 12B; see 3.2.2)** — or omitted for ventricular-like identity. Y-27632 was added at each transfer step to support cell survival. Spontaneous beating appeared by Day 6–8, and organoids were maintained in Advanced MEM (AMEM) with 1% penicillin–streptomycin and 1% GlutaMAX for long-term culture and maturation.

3.2.2 Three-Dimensional Aggregation Methods

Two aggregation platforms were used — ULA-plate centrifugation and orbital shaking — with the ULA platform run under two transfer schedules, yielding three protocols **(Fig. 12B)**. In both ULA protocols, dissociated cells seeded at 75,000 cells/well in round-bottom ultra-low-attachment 96-well plates (Corning, 7007) and centrifuged (200 × g, 3 min) to form uniform aggregates **(Supple Fig. S2).**

(a) Monolayer-to-ULA (delayed aggregation) **(Supple Fig. S4Aa, Ba).** Cells were differentiated as a 2D monolayer through the cardiomyocyte stage and transferred to ULA plates at Day 8; RA was applied from Day 3 to Day 5.

(b) Early ULA aggregation **(Supple Fig. S4Aa, Bb).** After 1.5 days of CHIR99021 treatment in 2D, cells were transferred to ULA plates at Day 1.5, with cardiac specification proceeding directly in 3D; RA was applied from Day 2 to Day 8.

(c) Shaker/Matrigel aggregation **(Supple Fig. S4Aa, Bc).** hiPSCs (~7×10^5 cells) were embedded in ~1% Matrigel in a 35 mm dish and cultured on an orbital shaker at 70 rpm from the outset; RA was applied from Day 2 to Day 5.

Morphological progression was monitored daily by brightfield microscopy, and chamber identity was validated at the protein level by immunofluorescence (Section 3.4).

3.3 Fabrication of 3D-Printed Micro-pillar Arrays and Pacing Platform

Micro-pillar arrays were designed in Rhinoceros 3D and sized to fit a standard 12-well plate, then fabricated on a high-resolution SLA printer (HALOT-MAGE PRO, Creality) using F39 White Flexible (rubber-like) photopolymer resin. Each unit comprised a 2 mm-thick circular base supporting nine cylindrical pillars (5 mm high) topped with 1 mm retaining prongs, defining a 3 mm inner space per well **(Fig 6)**. After printing, parts were washed in isopropyl alcohol to remove uncured resin, fully cross-linked in a UV post-curing chamber, sterilized in 70% ethanol, and UV-irradiated in a biosafety cabinet to minimize resin cytotoxicity.

For analysis, chamber-specific cardiac organoids (COs) were seated within the pillar array and embedded in Matrigel to hold them in position and provide an ECM-rich microenvironment **(Fig. 6A, right)**; the pillars served to position and stabilize each organoid for imaging and pacing, not to apply uniaxial tension.

For controlled pacing during electro-mechanical and optical analysis, a custom electrode system was integrated **(Fig. 6B, C)**: copper-wire electrodes were threaded through a PLA-printed holder, with their tips immersed in the culture medium on either side of the organoid, and connected to a field stimulator (MyoPacer, IonOptix, Westwood, MA, USA). Responses were evaluated under field stimulation at 1–3 Hz, 2 ms pulse, 5 V/cm, enabling standardized quantification of chamber-specific kinetics during imaging and Ca^{2+} mapping (Sections 3.7–3.8).

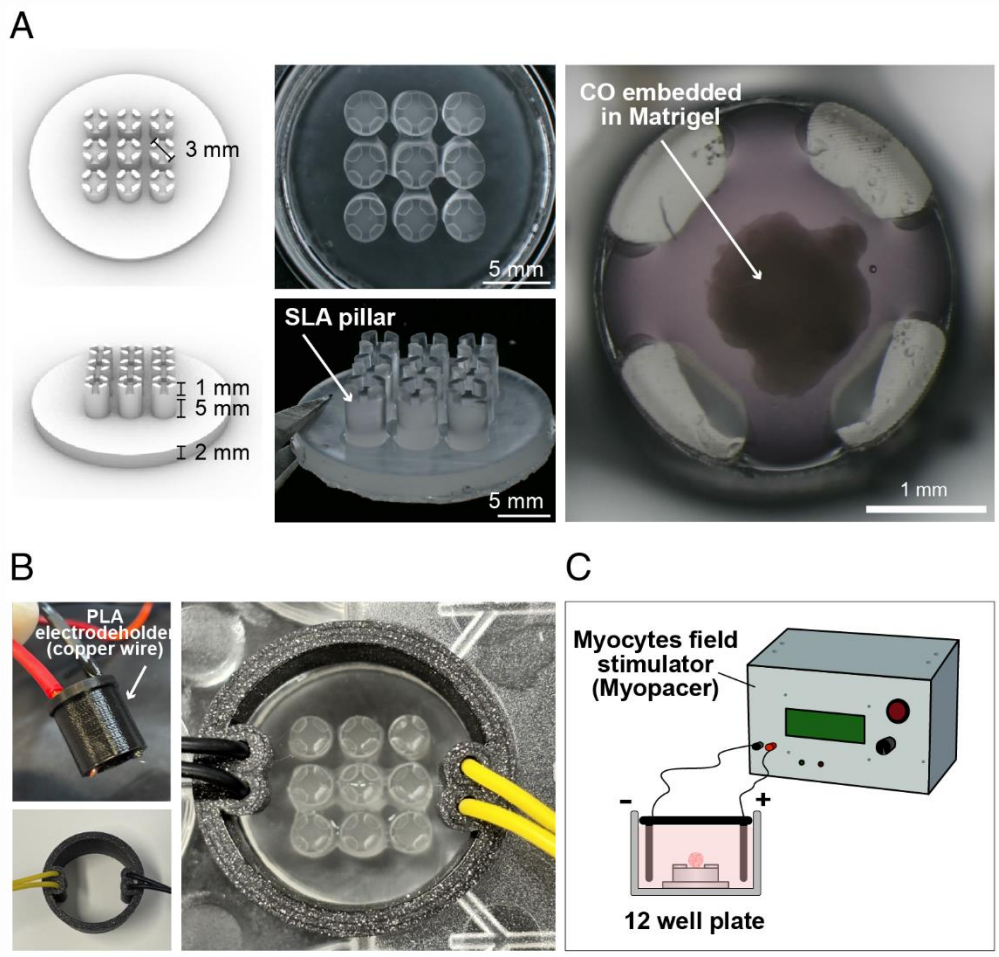


Fig. 6 Design and fabrication of the 3D-printed micro-pillar array and field-pacing platform. (A) Micro-pillar array design and organoid loading. *Left*, CAD renderings in top view (top) and side view (bottom): a 2 mm-thick circular base supporting nine cylindrical pillars (5 mm high) capped with 1 mm retaining prongs, defining a 3 mm inner space. *Middle*, photographs of the SLA-printed array (top) and a magnified single pillar unit (bottom, "SLA pillar"); scale bars, 5 mm. *Right*, representative top-down image of a chamber-specific cardiac organoid (CO) embedded in Matrigel and retained by the prongs; scale bar, 1 mm. **(B)** Custom field-pacing electrodes. *Left*, copper-wire electrodes threaded through a 3D-printed PLA holder (top) and the assembled electrode ring (bottom). *Right*, top-down view of the micro-pillar array seated in a culture well with the electrode ring positioned around the pillars. **(C)** Schematic of the pacing configuration: paired electrodes connected to a myocyte field stimulator (MyoPacer) deliver field stimulation (− / +) across the organoid in a 12-well plate

3.4 Immunofluorescence Staining

To assess structural morphology and chamber-specific identity of the 3D cardiac organoids (COs), a modified whole-mount immunofluorescence protocol (~6–7 days) was used to ensure antibody penetration into the thick tissue.

3.4.1 Fixation and Permeabilization

Organoids were fixed in 4% paraformaldehyde overnight at 4 °C, washed three times in PBS (15 min each), permeabilized in 0.1% Triton X-100/PBS for 3–4 h at room temperature, washed again (3 × 15 min), and blocked in 5% BSA/0.3% Triton X-100/PBS overnight at 4 °C.

3.4.2 Antibody Incubation and Background Reduction

Primary antibodies **(Table 2)** were diluted 1:200 in 1% BSA/0.1% Triton X-100/PBS — ~1.5–2× more concentrated than for 2D cultures, to ensure uniform labeling through the organoid core — and applied for 48–72 h at 4 °C with gentle agitation. Markers included structural (α-actinin/ACTN2, cardiac troponin T/cTnT; Figure 11) and chamber-identity proteins (MLC2a, MLC2v/MYL2, Cx43, Cx40; Figure 12). After extensive washing (0.1% Triton X-100/PBS; 5–6 × 1 h at RT, then overnight at 4 °C), Alexa Fluor Plus 488/647 secondary antibodies (1:500) and DAPI were applied for 24–48 h at 4 °C in the dark.

3.4.3 Mounting and Confocal Imaging

Organoids were washed (4–5 × 1 h in wash buffer, then 2 × 15 min in PBS), mounted in Fluoromount-G (Thermo Fisher, 00-4958-02) with custom spacers to preserve spherical morphology, and cured overnight at RT in the dark3D Z-stacks were acquired by confocal laser-scanning microscopy (K1-Fluo, Nanoscope Systems, Daejeon, South Korea) to resolve sarcomeric organization and the spatial distribution of chamber-specific proteins.

3.5 Gene Expression Analysis by qRT-PCR

To evaluate chamber-specific lineage specification of the 2D monolayer-derived cardiomyocytes at the transcriptional level, total RNA was extracted with the RNeasy

Mini Kit (Qiagen) according to the manufacturer's instructions, and RNA concentration and purity were measured by spectrophotometry. Complementary DNA (cDNA) was synthesized from 500 ng total RNA using AccuPower RT MasterMix (Bioneer) with Oligo-dT_{20} primers (Bioneer) and nuclease-free water. Quantitative real-time PCR was performed with PowerUp SYBR Green Master Mix (Thermo Fisher Scientific) and target-specific primers **(Table 3)** in MicroAmp Optical 96-well reaction plates (0.2 mL; Applied Biosystems, N8010560) on a QuantStudio 1 Real-Time PCR System (96-well, 0.2 mL; Applied Biosystems). Cycling conditions were 50 °C for 2 min and 95 °C for 2 min, followed by 40 cycles of 95 °C for 15 s and 60 °C for 60 s, with a final melt-curve analysis. Relative expression of pan-cardiac markers (NKX2-5, TNNT2), the atrial marker NR2F2, and the ventricular marker IRX4 was calculated by the comparative 2^(−ΔΔCt) method[44] and normalized to GAPDH.

3.6 Flow Cytometry for Cardiomyocyte Purity

To quantify cardiomyocyte differentiation efficiency, intracellular flow cytometry for cardiac troponin T (cTnT) was performed on the 2D monolayer-derived cardiomyocytes (Figure 9C). Cells (Day 15–20, post-metabolic selection) were dissociated with TrypLE Select (Gibco) for 5–10 min at 37 °C, neutralized with culture medium, and centrifuged (300 × g, 5 min). The pellet was washed once in PBS, fixed in 4% PFA for 15 min at room temperature, and permeabilized with 0.1% saponin/PBS for 15 min. Cells were stained with a FITC-conjugated anti-cTnT antibody (REAfinity; Miltenyi Biotec, 130-119-575; 1:100) for 30 min at 4 °C in the dark, with an unstained control prepared in parallel for gating. After two washes in 0.1% saponin/PBS, cells were resuspended in PBS and analyzed on a BD FACSCanto II flow cytometer (BD Biosciences). The $cTnT^+$ population was defined relative to the unstained control, and data were analyzed in FlowJo (BD Biosciences).

3.7 Optical Imaging for Ca^{2+} Transient Recording

For Ca^{2+}-transient imaging, cardiac organoids were loaded with the cell-permeable Ca^{2+} indicator Fluo-4 AM (FluoroPure grade; Invitrogen/Thermo Fisher Scientific, F23917; 5 μM final) in Tyrode's solution for 30–40 min at 37 °C, followed by a 15 min de-esterification wash in dye-free Tyrode's solution. Loaded organoids embedded in the micro-pillar arrays (Section 3.3) were imaged at 37 °C on a Nautilai multiwell fluorescence imaging system (Curi Bio, Seattle, WA, USA). Field pacing was delivered via the MyoPacer stimulator at 1–3 Hz (5 V/cm, 2 ms pulse). Recordings were 10–30 s per ROI, with longer acquisitions for activation mapping (up to 7,500 frames; 30 s at 250 fps). Spontaneous activity was recorded immediately before pacing. Acquired image sequences were exported and analyzed with the custom iCAMAnalyzer platform (Section 3.8).

3.8 iCAMAnalyzer: Unified Python-Based Analytical Platform for Electro-Mechanical Quantification

iCAMAnalyzer (iPSC-CO Calcium & Motion Analysis) is a custom platform developed entirely in Python with a PyQt5 graphical user interface (GUI). It consolidates two functional readouts of the engineered cardiac organoids into one standardized workflow. At launch, the GUI prompts the user to select the video type **(Fig. 7)**: **Fluorescence**, which routes the recording to Ca^{2+}-transient kinetics and spatiotemporal activation mapping, or **Brightfield**, which routes it to motion and contractility analysis. Crucially, the two pipelines operate on independent recordings — Fluo-4 fluorescence videos (acquired on the Nautilai system, Section 3.7) for Ca^{2+} and label-free brightfield videos for motion, not the same video. Both branches share common signal-processing routines and export standardized spreadsheets (openpyxl) and figures (matplotlib) for algorithmic consistency across batches.

3.8.1 Software architecture and workflow

Once the video type is confirmed, the corresponding module loads: the Fluorescence branch runs (i) Ca^{2+}-transient kinetics and (ii) activation mapping, whereas the label-

free branch runs multi-organoid motion/contractility tracking. All branches share ROI handling, filtering, peak detection, and export routines.

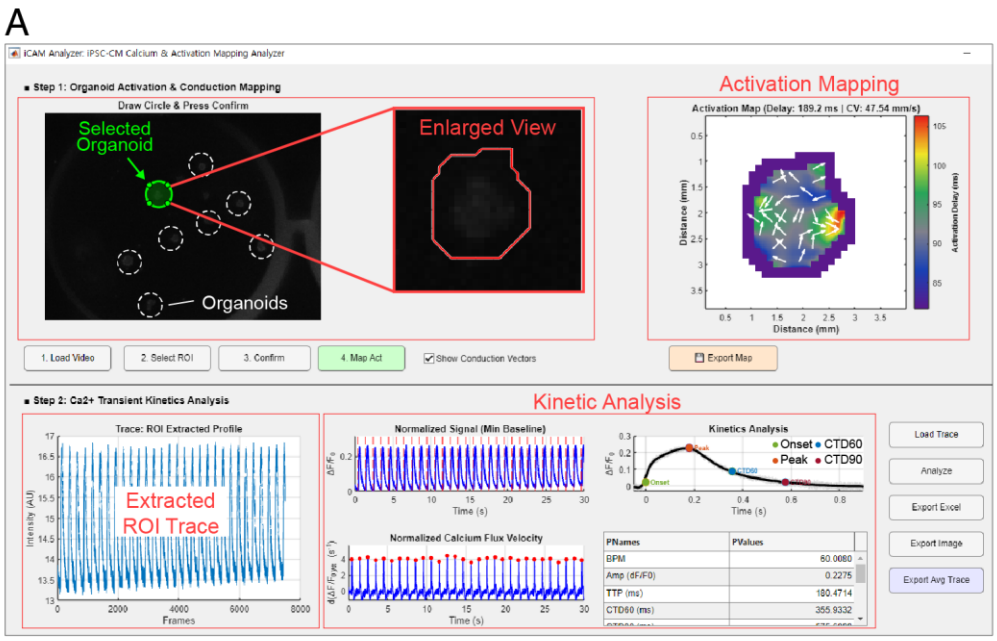
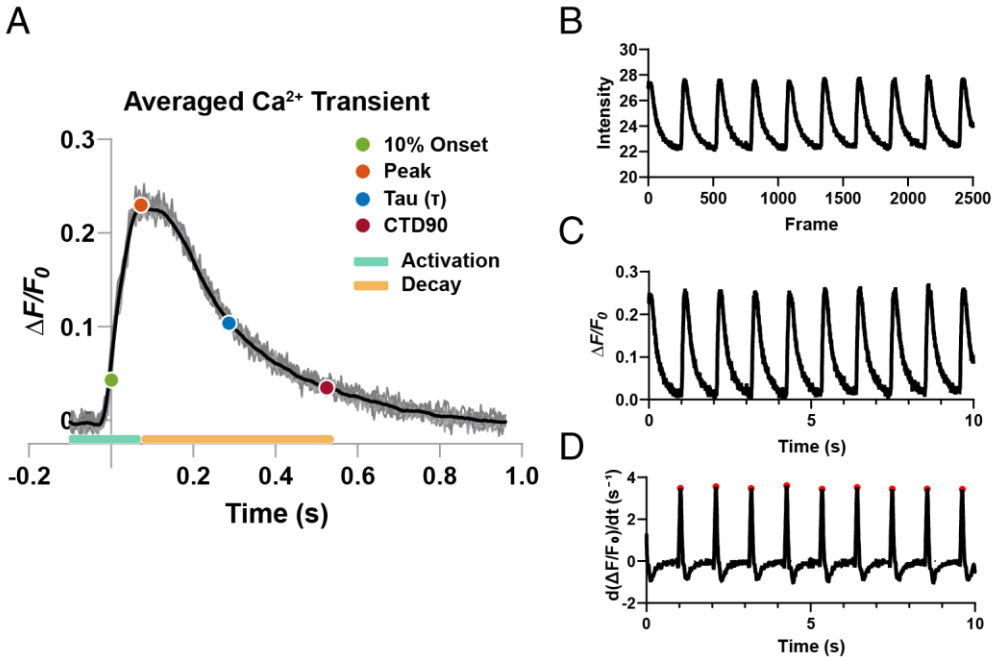


Fig. 7 Fluorescence-mode interface of iCAMAnalyzer for Ca^{2+} transient kinetics and spatiotemporal activation mapping. A representative overview of the custom-developed Python-based graphical user interface (GUI), "iCAMAnalyzer," designed for the high-throughput evaluation of engineered cardiac tissues. The system is bifurcated into two specialized functional modules: (Top Red Box) Step 1. Spatial Activation & Conduction Mapping: This module provides tools for spatial analysis, including region-of-interest (ROI) definition, activity-based cellular masking, and the generation of high-resolution activation maps. It allows for the precise calculation of macroscopic conduction velocity (CV) by spatializing the temporal delay of calcium waves across the organoid surface. (Bottom Red Box) Step 2. Ca2+ Transient Kinetics Analysis: This module focuses on temporal dynamics by extracting raw fluorescence intensity traces from defined ROIs. The system automatically performs signal normalization ($\Delta F/F_0$) and baseline correction, followed by the derivation of key kinetic parameters such as time-to-peak (TTP), decay time constant (τ), and transient durations through ensemble averaging of multiple consecutive beats

3.8.2 Fluorescence Ca^{2+} transient kinetics

Raw intensity is extracted from a user-defined ROI, smoothed with a Savitzky–Golay filter, and normalized to $\Delta F/F_0$ using a baseline F_0 defined as the mean of the lowest 10%

of the trace. Beats are detected on the smoothed first derivative (dF/dt) with an FFT-adaptive peak-distance (dominant frequency searched over 0.5–4.0 Hz; minimum distance ≈65% of the estimated period). Per beat, the platform computes beating rate (BPM), amplitude ($\Delta F/F_0$), time-to-peak (TTP), CTD_{60}/CTD_{90}, decay constant (τ, mono-exponential fit), and upstroke/decay velocities (Vmax, Vrelax) **(Fig. 8).** To quantitatively distinguish chamber identity, two normalized indices are derived (Figure 7/8): a **Shape Index (TTD/CTD_{90})**, capturing plateau-like (ventricular, high) versus spike-like (atrial, low) morphology, and an **Efficiency Index (Vmax/amplitude)**, reflecting amplitude-normalized kinetics (ventricular low, atrial high).

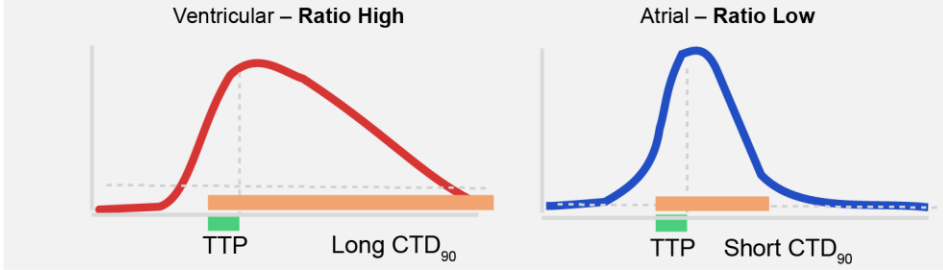


E

Chamber Specificity Indices

Shape Index **TTD/CTD$_{90}$**

Activation -to-relaxation morphology - plateau vs spike



Efficiency Index **Vmax/Amp**

Amplitude-normalized Ca2+ rise & decay velocity

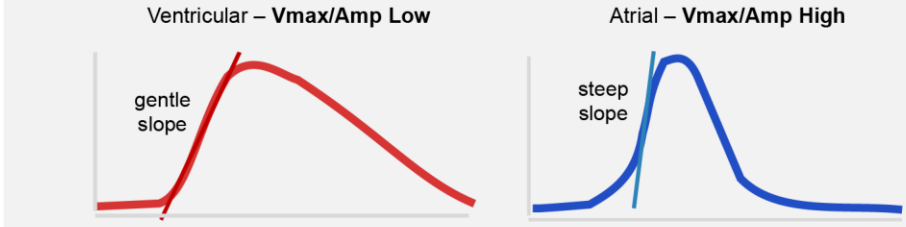


Fig. 8 Automated calcium transient kinetic analysis and parameter extraction workflow. (A) Representative user interface of the custom Python-based software for temporal kinetics analysis. The module automatically extracts raw fluorescence traces from the selected region of interest (ROI), normalizes the signal, and computes the first derivative to evaluate real-time calcium flux velocity. **(B)** Detailed methodology for the quantitative evaluation of calcium kinetics. (Left) The raw fluorescence signal is normalized $\Delta F/F_0$, and its first derivative (dF/dt) is computed in parallel. The red arrow highlights the exact temporal synchronization between the maximum positive peak of the derivative and the steepest activation phase of the transient. (Right) Multiple consecutive beats are synchronized and overlaid via ensemble averaging to generate a robust, representative "Average Transient." Key kinetic parameters, including Time to Peak (TTP) for the activation phase, as well as the decay time constant (tau) and 90% calcium transient duration (CTD90) for the decay phase, are precisely derived from this master trace.

3.8.3 Spatiotemporal activation mapping

For fluorescence stacks, the local activation time (T) at each active pixel is defined as the 50%-upstroke crossing within an automated, beat-period-scaled window. Macroscopic conduction velocity (CV = $1/\|\nabla T\|$) is extracted by a two-pass adaptive Bayly gradient method: a first pass estimates a rough CV, and the analysis window is then adaptively scaled to the local CV (≈7×7 up to 13×13 pixels) — widening where inter-pixel delays approach the frame interval and gradient noise grows. An interquartile-range (IQR) fence removes outlier gradients before the median CV is reported, and conduction vectors are overlaid on an isochronal map **(Fig. 7; Supple Fig. S8).**

3.8.4 Label-free motion and contractility

For label-free dark-field videos, up to nine organoids per field are segmented automatically (adaptive Otsu thresholding + contour filtering by area, solidity, and aspect ratio). For each ROI, frame-to-frame absolute-difference mapping (cv2.absdiff) against an adaptively updated diastolic reference yields a displacement trace, which is linearly detrended; contraction widths are measured at 90% amplitude and mechanical output is reported as percentage Fractional Area Change (FAC %), analogous to ejection fraction **(Fig. 9).**

A

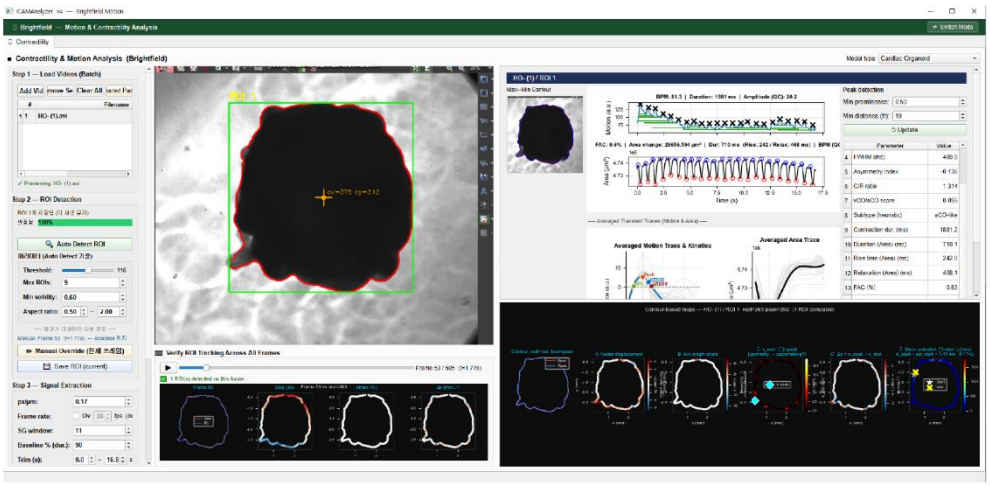


Fig. 9 Automated mechanical motion and contractility analysis. Intrinsic optical signal-based mechanical analysis captures both global organoid area changes and localized ROI dynamics, offering an integrated view of whole-tissue and regional behavior.

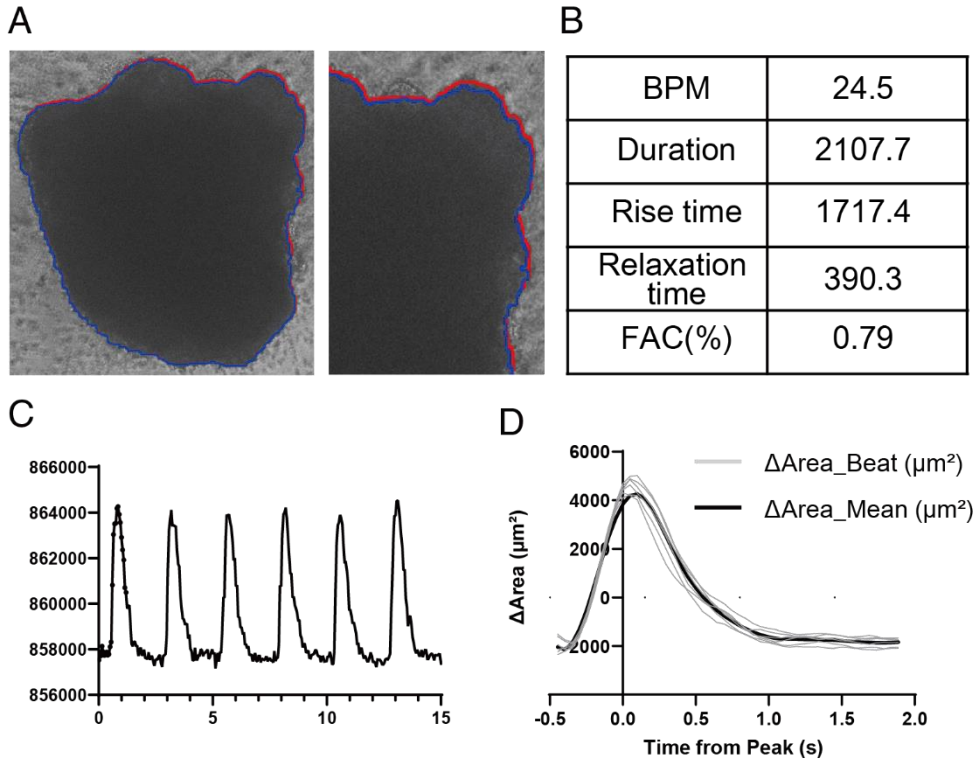


Fig. 10 Label-free quantification of contractile function by automated boundary tracking. (A) Representative brightfield images of a chamber-specific cardiac organoid with the automatically detected tissue boundary overlaid; this contour is tracked frame-by-frame to quantify the change in projected area between relaxation and contraction. (B) Representative single-organoid summary of the extracted contractile parameters: beating rate (BPM), contraction duration, rise time (time-to-peak), relaxation time, and Fractional Area Change (FAC, %). (C) Raw projected-area trace over time, from which individual contraction cycles are automatically detected. (D) Beat-to-beat ΔArea transients aligned to their peaks (gray, ΔArea_Beat) overlaid with their ensemble average (black, ΔArea_Mean), yielding a representative contraction waveform for parameter extraction

3.9 Pharmacological Treatment

To probe chamber-specific functional responses, organoids were challenged with two agents: carbachol (CCh), a muscarinic agonist that activates the atrial-selective acetylcholine-gated K^+ current ($I_{K,ACh}$), and isoproterenol (ISO), a non-selective β-adrenergic agonist used as a general chronotropic/inotropic stimulus. Carbachol (Sigma-Aldrich, C4382) was prepared as a 10 mM stock in distilled water; isoproterenol (Sigma-Aldrich, I5627) was prepared as a 10 mM stock in distilled water containing 1 mM ascorbic acid (antioxidant, owing to its rapid oxidation). Both were diluted to 1 μM in culture medium immediately before use. For each experiment, baseline Ca^{2+} transients and/or contractions were recorded under 1 Hz pacing for 10 s. The medium was then replaced with drug-containing medium, and organoids were equilibrated for 10 min at 37 °C before a post-treatment recording under identical conditions. Pre- vs post-treatment data from the same organoid were compared by paired t-test (Section 3.12).

3.10 Cell Viability Assessment via Live/Dead Assay

Viability and cytotoxicity of iPSC-CMs and cardiac organoids were assessed by a dual-fluorescence Live/Dead assay using Calcein-AM (live; Invitrogen, C3100MP) and ethidium homodimer-1 (EthD-1; dead; Invitrogen, E1169). A working solution was freshly prepared immediately before use by diluting the stocks in serum-free Tyrode's solution (pH 7.4) to 2 μM Calcein-AM and 4 μM EthD-1; Tyrode's solution was used to prevent premature esterase cleavage of Calcein-AM by serum and to avoid phenol-red background. After aspirating the medium and washing once with Tyrode's, samples were submerged in the working solution and incubated for 30–45 min (60–90 min for 3D organoids) at 37 °C in the dark. Samples were imaged directly (without removing the staining solution) on a confocal laser-scanning microscope (K1-Fluo, Nanoscope Systems, Daejeon, South Korea): calcein via a FITC/GFP filter (ex/em 494/517 nm) and EthD-1 via a Texas Red filter (ex/em 528/617 nm). Merged images were analyzed in

ImageJ (NIH), and viability was reported as the ratio of live to total cells **(Supple Fig. S5).**

3.11 Statistical Analysis

All quantitative data are presented as mean ± SEM unless otherwise stated in the figure legends, and all analyses were performed in GraphPad Prism 8.0.2 (GraphPad Software, San Diego, CA, USA). The number of replicates (n) denotes independent biological replicates — individual organoids or differentiation runs — as specified in each figure legend; when multiple beats were recorded from a single organoid, they were averaged to one per-organoid value before analysis to avoid pseudoreplication.

Two independent groups (e.g., ventricular vs atrial organoids) were compared by a two-tailed unpaired t-test with Welch's correction when data were normally distributed (equal variances not assumed), or by the non-parametric Mann–Whitney U test when the normality assumption was not met. Paired measurements from the same sample before and after treatment (pharmacology, Section 3.9) were compared by a two-tailed paired t-test. Multiple independent groups (e.g., viability or morphology across seeding densities) were compared by one-way ANOVA followed by Tukey's (all-pairwise) or Dunnett's (versus a single control) post-hoc test, with homogeneity of variance verified by the Brown–Forsythe and Bartlett's tests.

$p < 0.05$ was considered statistically significant (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; ns, not significant).

Table 2. List of antibodies and fluorescent stains used for immunofluorescence

Target / Reagent	Manufacturer	Catalog Number
Primary Antibodies		
alpha-Actinin (ACTN2)	Sigma-Aldrich	A7811-100UL
Cardiac Troponin T (cTnT)	Thermo Fisher Scientific	MA512960
MLC2a	Synaptic Systems	311 011
MLC2v (Myosin Light Chain 2)	Proteintech	10906-1-AP
Connexin 43 (Cx43)	Cell Signaling Technology	3512
Connexin 40 (Cx40)	Santa Cruz Biotechnology	sc-365107
Secondary Antibodies		
Alexa Fluor™ Plus 488	Thermo Fisher Scientific	A32731
Alexa Fluor™ Plus 647	Thermo Fisher Scientific	A32728
Fluorescent Stains		
Phalloidin (F-actin)	Thermo Fisher Scientific	A12379
DAPI	Sigma-Aldrich	D5942

Table 3. Primer sequences used for qPCR

Gene	Forward Primer	Reverse Primer
Housekeeping		
GAPDH	ATGTTCGTCATGGGTGTGAA	TGAGTCCTTCCACGATACCA
Cardiac Progenitor & Pan-Cardiac		
NKX2-5	GGCCCATACTCCATGGTTTA	CACAGGAGGTGACGGAGTGT
TNNT2	GACAGAGGCAAAGTGGGAAG	TGAAGGAAGGCCAGGCTCTAT
Atrial Specific		
NR2F2	CCAGCGGGTGGGTCGCCTTTATGGA	CGGGCTGTTGGGGTGCAGGTCCCTA
Ventricular Specific		
IRX4	TTCCGTTCTGAAGCGTGGTC	TGAAGCAGGCAAGATTTGGTGT

IV.Results

4.1 An Integrated Optical Platform for Electro-Mechanical Analysis

All functional readouts in this study were acquired and analyzed through iCAMAnalyzer (iPSC-CO Calcium & Motion Analysis), a custom, unified platform developed entirely in Python **(Fig. 7-10)**. At launch, the user selects the recording modality: the Fluorescence mode extracts Ca^{2+}-transient kinetics and reconstructs spatiotemporal activation maps **(Fig. 7,8)**, while the Brightfield mode performs label-free motion and contractility analysis **(Fig. 9, 10)**. The two modes are applied to the same organoids but to independent recordings — Fluo-4 fluorescence videos for Ca^{2+} and brightfield videos for motion — with all parameters extracted from a user-defined ROI and summarized in Table 4. Optical acquisition was performed under field pacing delivered through the 3D-printed micro-pillar/electrode platform **(Fig. 6)**. The following sections apply this platform to the structural, molecular, electrophysiological, mechanical, and pharmacological characterization of chamber-specific 3D cardiac organoids (vCOs and aCOs).

4.2 Efficient Generation and Validation of Chamber-Specific Cardiomyocytes

hiPSCs were directed toward atrial- and ventricular-like fates by stage-specific modulation of Wnt signaling, using CHIR99021 for early activation followed by IWR-1 for inhibition. Chamber identity was determined at the cardiac mesoderm stage: atrial specification was driven by retinoic acid (RA) supplementation, whereas ventricular specification was achieved by RA omission **(Fig. 11A).**

RT-qPCR confirmed efficient and comparable cardiac induction across both lineages. Expression of the pan-cardiac markers TNNT2 and NKX2-5 did not differ significantly between groups (ns), indicating equivalent myogenic differentiation. In contrast, chamber-identity markers segregated sharply: the atrial determinant NR2F2 was strongly enriched in atrial-like cells (approximately 14-fold over ventricular; ***p <

0.001), while the ventricular determinant IRX4 was expressed almost exclusively in ventricular-like cells (***$p < 0.001$) **(Fig. 11B).** Intracellular flow cytometry for cardiac troponin T (cTnT) further confirmed a high-purity cardiomyocyte population. Together, these data establish that the protocol yields pure, lineage-resolved chamber-specific cardiomyocytes at the monolayer stage **(Fig. 11C).**

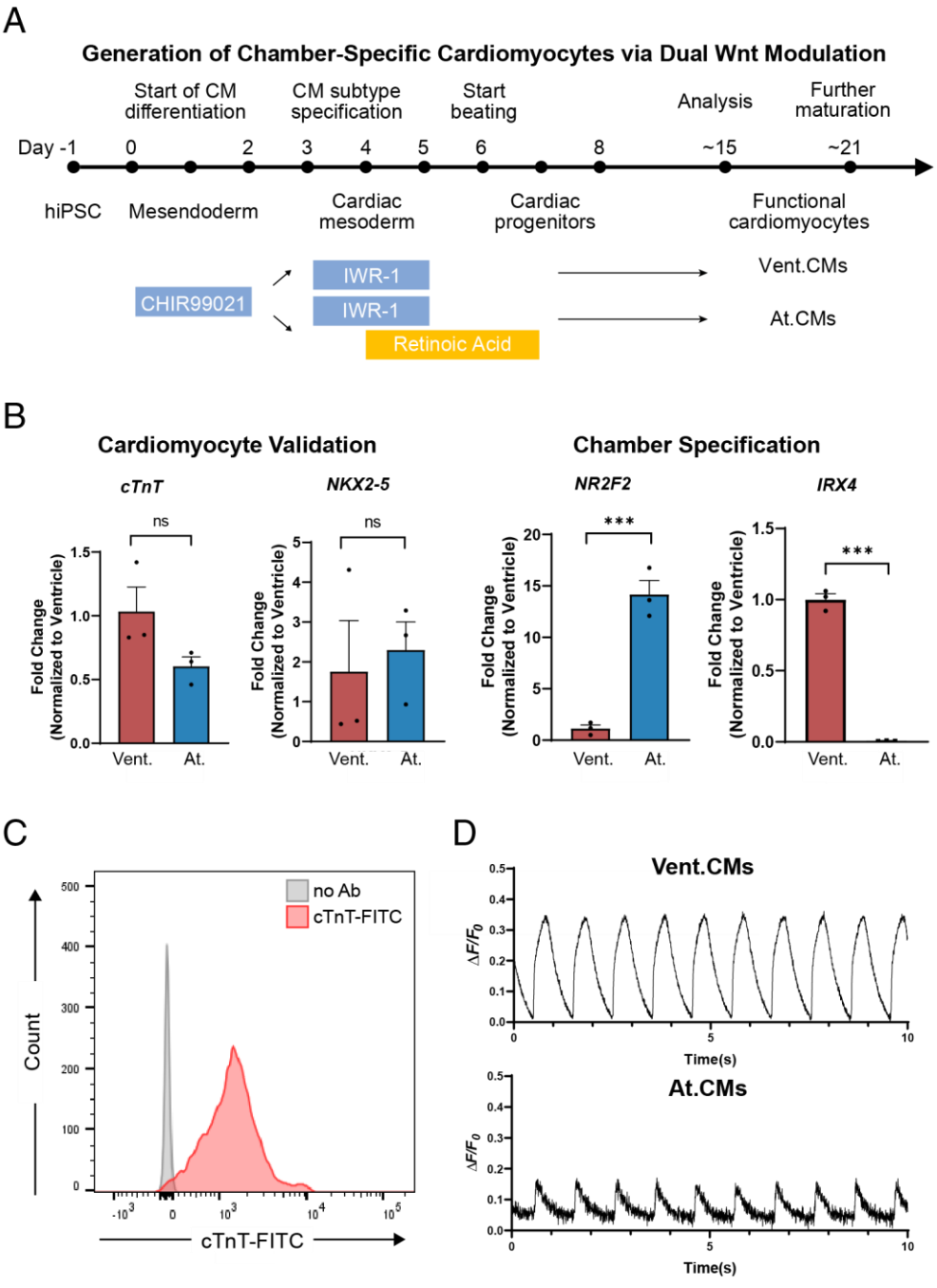

A
Generation of Chamber-Specific Cardiomyocytes via Dual Wnt Modulation
Start of CM differentiation
CM subtype specification
Start beating
Analysis
Further maturation
Day -1
0
2
3
4
5
6
8
~15
~21
hiPSC
Mesendoderm
Cardiac mesoderm
Cardiac progenitors
Functional cardiomyocytes
CHIR99021
IWR-1
IWR-1
Retinoic Acid
Vent.CMs
At.CMs
B
Cardiomyocyte Validation
Chamber Specification
cTnT
NKX2-5
NR2F2
IRX4
ns
ns
***
***
Fold Change (Normalized to Ventricle)
Vent.
At.
C
no Ab
cTnT-FITC
Count
cTnT-FITC
D
Vent.CMs
At.CMs
ΔF/F0
Time(s)

Fig. 11 Efficient generation and validation of chamber-specific cardiomyocytes. (A) Schematic timeline of the directed differentiation protocol. hiPSCs were specified toward ventricular- or atrial-like fates by stage-specific Wnt modulation (CHIR99021 → IWR-1); atrial fate was induced by retinoic acid (RA) during the cardiac mesoderm stage. (B) RT-qPCR validation. Pan-cardiac markers (TNNT2, NKX2-5) showed no significant difference between groups, confirming consistent myogenic induction, whereas chamber markers were lineage-restricted: NR2F2 was up-regulated in atrial-like cells and IRX4 expressed exclusively in ventricular-like cells. Cardiomyocyte purity was confirmed by cTnT flow cytometry. Data are mean ± SEM (n = 3); Student's t-test (***$p < 0.001$; ns, not significant).

4.3 Biphasic Differentiation and Morphological Evolution of 3D Cardiac Organoids

To recapitulate the structural hierarchy of the native myocardium, differentiating cells were transitioned from a 2D monolayer—used to ensure uniform mesendoderm induction—into a 3D suspension culture. After 1.5 days of CHIR99021 treatment, cells were dissociated, seeded into Ultra-Low Attachment (ULA) 96-well plates, and aggregated by brief centrifugation (180 × g, 3 min). Chamber identity was subsequently specified within the 3D aggregates by IWR-1 treatment with or without RA **(Fig. 12A).** Brightfield monitoring documented progressive self-organization. Loosely packed aggregates compacted into uniform, smooth-bordered spheroids over the first days of culture, and both ventricular (vCO) and atrial (aCO) organoids initiated spontaneous macroscopic contraction around Day 6 **(Fig. 12B).** Both lineages followed comparable morphological trajectories, indicating that the biphasic protocol supports robust and reproducible organoid formation independent of chamber fate.

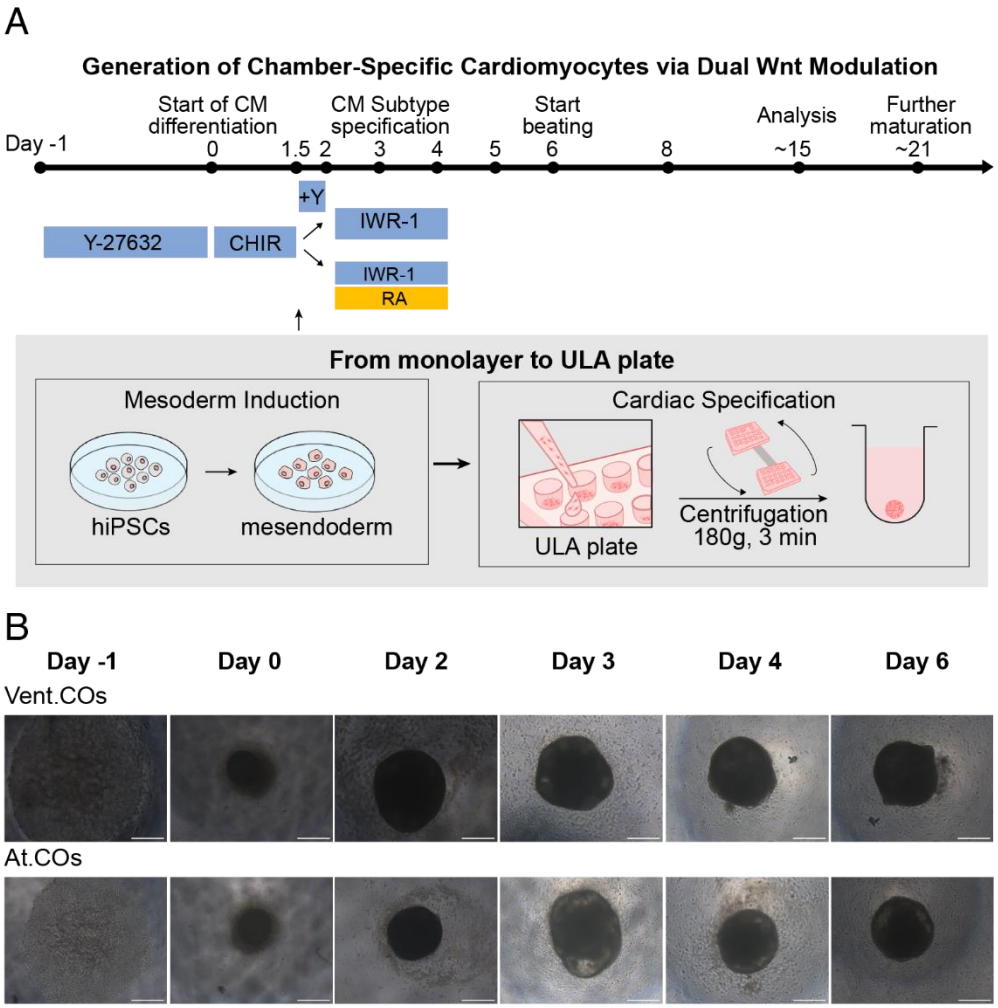


Fig. 12 Biphasic differentiation protocol and morphological evolution of chamber-specific 3D cardiac organoids. (A) Biphasic workflow. Mesoderm induction is performed in a 2D monolayer before cells are transferred to ULA 96-well plates and aggregated by centrifugation (180 × g, 3 min); chamber identity is then specified in the 3D aggregates by IWR-1 ± RA. (B) Representative brightfield images tracking organoid formation and compaction for ventricular and atrial lineages. Both organoid types form uniform spheroids and begin spontaneous beating around Day 6. Scale bars: 500 µm.

4.4 Morphological Characterization and Sarcomeric Maturation

Whole-mount confocal imaging of cardiac troponin T (cTnT) revealed dense, highly interconnected myofibrillar meshworks in both organoid types **(Fig. 13A).** Notably, vCOs reproducibly developed an internal chamber-like cavity, a structural hallmark of ventricular morphogenesis that was not a feature of aCOs.

High-resolution α-actinin imaging resolved individual Z-disc registers and revealed chamber-dependent sarcomeric maturation **(Fig. 13B).** vCOs exhibited well-aligned, anisotropic sarcomeres with an elongated sarcomere length (SL) of approximately 2.1 µm, approaching the adult ventricular range, whereas aCOs displayed shorter, less-ordered sarcomeres with an SL of approximately 1.7 µm. The greater sarcomere length and alignment in vCOs is consistent with the higher contractile demand of the ventricular lineage.

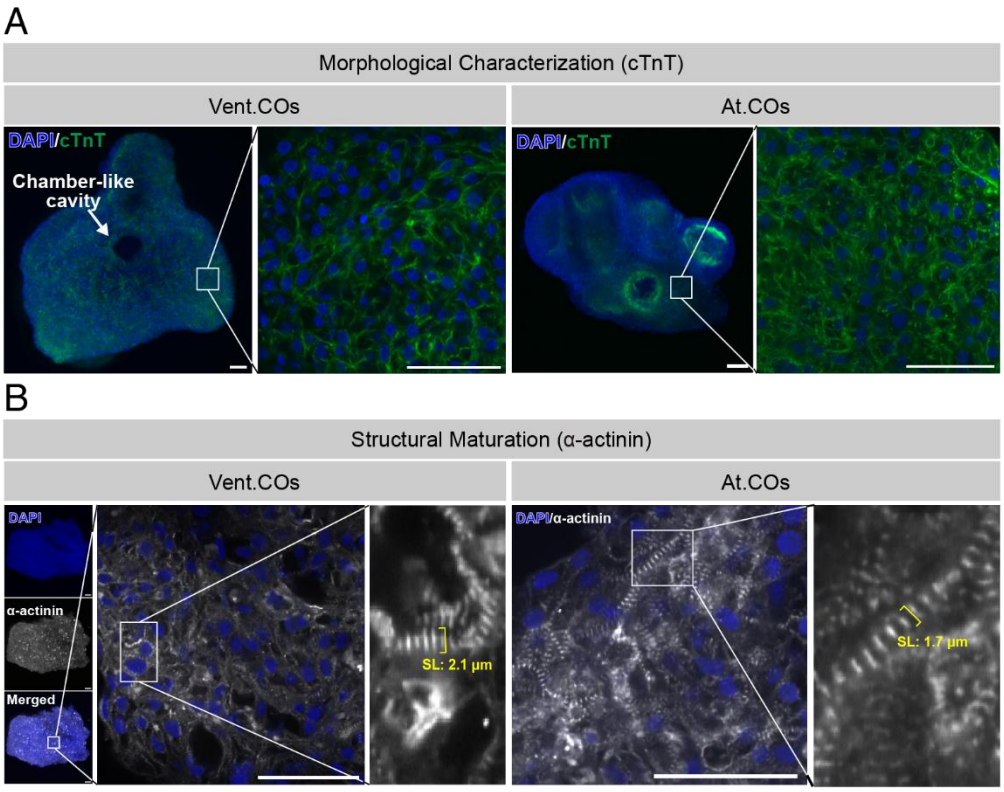


Fig. 13 Morphological characterization and sarcomeric maturation of chamber-specific 3D cardiac organoids. (A) Confocal overviews and high-magnification wall insets of ventricular (Vent.COs) and atrial (At.COs) organoids stained for cTnT (green). Ventricular organoids display a distinct central chamber-like cavity (arrow); both lineages form dense cTnT-positive myofibrillar networks. (B) High-resolution α-actinin imaging (grey) resolving Z-disc alignment. Ventricular organoids reach an elongated sarcomere length (SL ≈ 2.1 µm), whereas atrial organoids show shorter spacing (SL ≈ 1.7 µm). Nuclei, DAPI (blue). Scale bars: 500 µm.

4.5 Molecular Identity and Gap Junction Networking

Chamber-specific sarcomeric isoform expression confirmed lineage identity at the protein level **(Fig. 14A).** vCOs robustly and selectively expressed the ventricular isoform MLC2v, whereas aCOs were dominated by the atrial isoform MLC2a; high-magnification insets resolved organized striations in both, confirming high-fidelity, lineage-restricted maturation. Gap junction profiling revealed chamber-appropriate connexin networks **(Fig. 14B).** Punctate Cx43 was distributed at cell–cell junctions throughout the ventricular tissue, while aCOs additionally expressed the atrial-enriched Cx40. The distinct, junctional localization of these connexins indicates the formation of a coupled, syncytial electrical network in both organoid types.

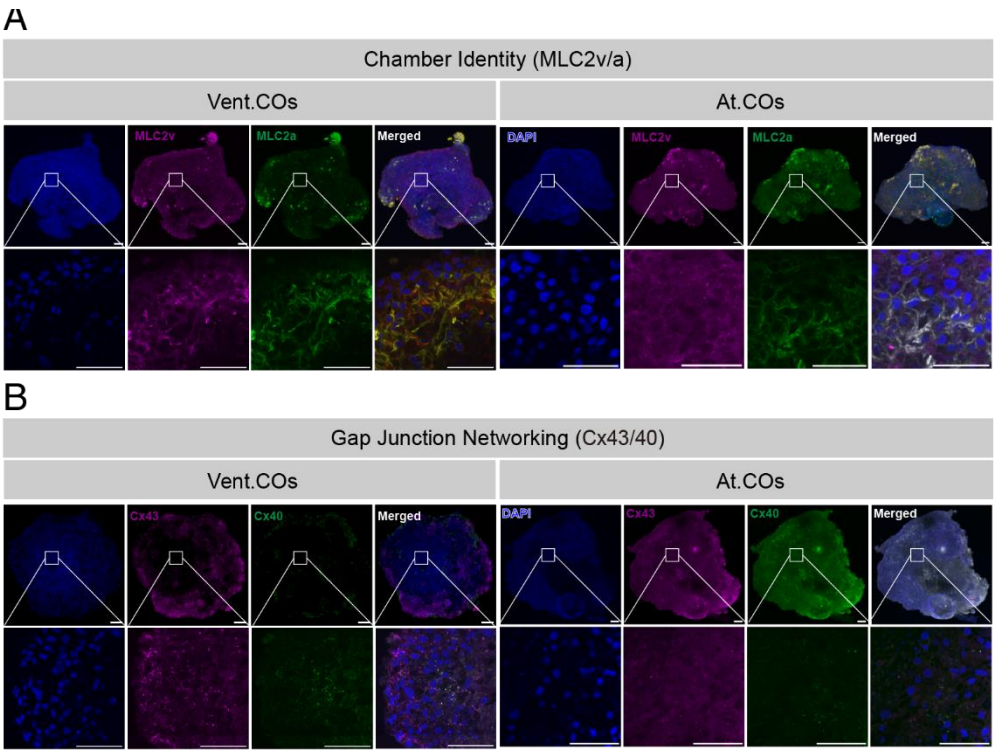


Fig. 14 Molecular identity and gap junction networking of chamber-specific 3D cardiac organoids. (A) Chamber-specific sarcomeric isoforms. Immunofluorescence for MLC2v (magenta) and MLC2a (green) with DAPI (blue). Ventricular organoids exclusively express MLC2v, whereas atrial organoids are dominated by MLC2a; insets resolve organized striations. (B) Gap junction networking. Cx43 (magenta) is distributed throughout ventricular tissue, while atrial organoids additionally express Cx40 (green); punctate junctional localization indicates a syncytial electrical network. Scale bars: 500 µm.

4.6 Dynamic Calcium Transient Profiling and Functional Fingerprinting

To quantify chamber-specific excitation kinetics, Ca^{2+} transients were analyzed with iCAMAnalyzer and condensed into two interpretable indices **(Fig. 15A–B).** vCOs produced broad, plateau-shaped transients, whereas aCOs produced sharp, spike-shaped transients.

The Shape Index (TTP/CTD90), which reflects activation-to-relaxation morphology, was significantly higher in vCOs than in aCOs (***$p < 0.001$), quantifying the ventricular plateau versus the atrial spike (Figure 13C, left). The Efficiency Indices captured the underlying flux dynamics: the normalized rise velocity (Vmax/Amp), reflecting the rate of Ca^{2+}-induced Ca^{2+} release, was markedly higher in aCOs (***$p < 0.001$), and the normalized decay velocity (Vrelax/Amp), reflecting the rate of SR Ca^{2+} re-uptake, was likewise higher in aCOs (***$p < 0.001$) (Figure 13C, middle and right). Together, these indices define a quantitative functional fingerprint: fast-cycling, spike-like atrial kinetics versus slow, sustained ventricular kinetics.

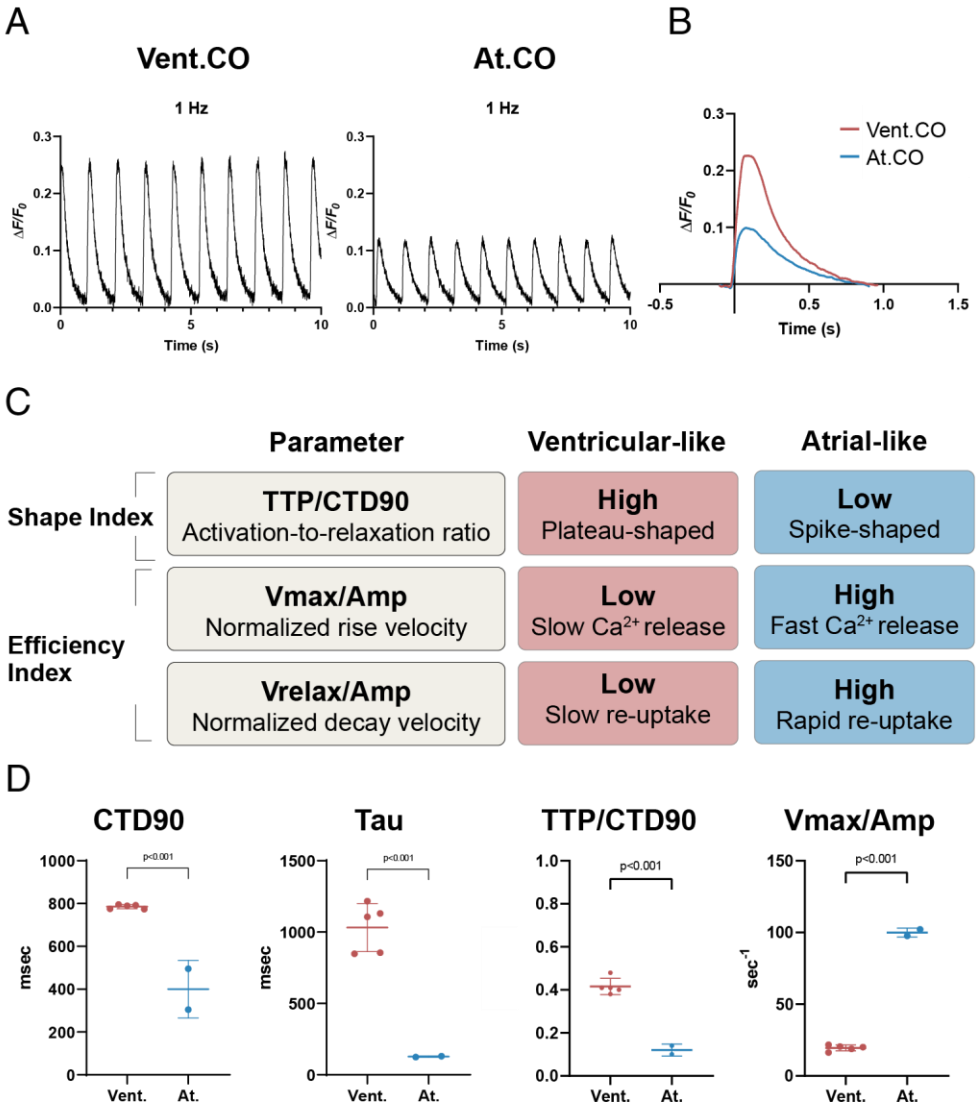

A
Vent.CO
At.CO
1 Hz
1 Hz
ΔF/F0
Time (s)
B
Vent.CO
At.CO
ΔF/F0
Time (s)
C
Parameter
Ventricular-like
Atrial-like
Shape Index
TTP/CTD90
Activation-to-relaxation ratio
High
Plateau-shaped
Low
Spike-shaped
Efficiency Index
Vmax/Amp
Normalized rise velocity
Low
Slow Ca2+ release
High
Fast Ca2+ release
Vrelax/Amp
Normalized decay velocity
Low
Slow re-uptake
High
Rapid re-uptake
D
CTD90
Tau
TTP/CTD90
Vmax/Amp
msec
msec
sec-1
p<0.001
Vent.
At.

Fig. 15 Dynamic calcium transient profiling and functional index formulation for chamber-specific 3D cardiac organoids. (A) Representative continuous calcium transient recordings. Real-time intracellular calcium ($\Delta F/F_0$) traces of ventricular (Vent.CO) and atrial (At.CO) organoids under a synchronized baseline pacing frequency of 1 Hz. (B) Single-beat ensemble overlay. Comparative trace overlay of a representative single calcium transient cycle between Vent.CO (red) and At.CO (blue), demonstrating distinct repolarization and decay slopes. (C) Algorithmic classification criteria for Shape and Efficiency Indices. Schematic table defining the customized functional matrix parameters used to mathematically distinguish ventricular-like (plateau-shaped, slow kinetics) versus atrial-like (spike-shaped, rapid kinetics) calcium fingerprints. (D) Quantitative evaluation of baseline calcium kinetics. Statistical comparisons of major electrophysiological parameters: CTD90, Tau (τ), TTP/CTD90 Ratio, and Vmax/Amp between the ventricular (Vent.) and atrial (At.) groups. Atrial organoids display accelerated activation and extraction kinetics across all indices. Data are presented as scatter dot plots showing individual data points with mean ± SEM; $p < 0.001$ by Student's t-test.

4.7 Spatiotemporal Conduction and Electrical Pacing Fidelity

The activation-mapping module of iCAMAnalyzer reconstructed Ca^{2+} wave propagation across the organoid surface **(Fig. 16, top).** Isochronal maps with overlaid conduction vectors revealed organized, directional propagation in both chamber types, with comparable macroscopic conduction velocities (approximately 33–35 mm/s; vCO ≈ 33.3 mm/s, aCO ≈ 34.6 mm/s).

Under incremental field pacing (1, 2, and 3 Hz), organoids were tested for stimulus-following fidelity **(Fig. 16, bottom).** vCOs maintained tight 1:1 capture across the full frequency range, tracking the pacing rate along the line of identity, whereas aCOs followed pacing at lower frequencies but showed greater variability and reduced capture at 3 Hz. This robust high-frequency entrainment confirms mature electrical coupling within the 3D organoids.

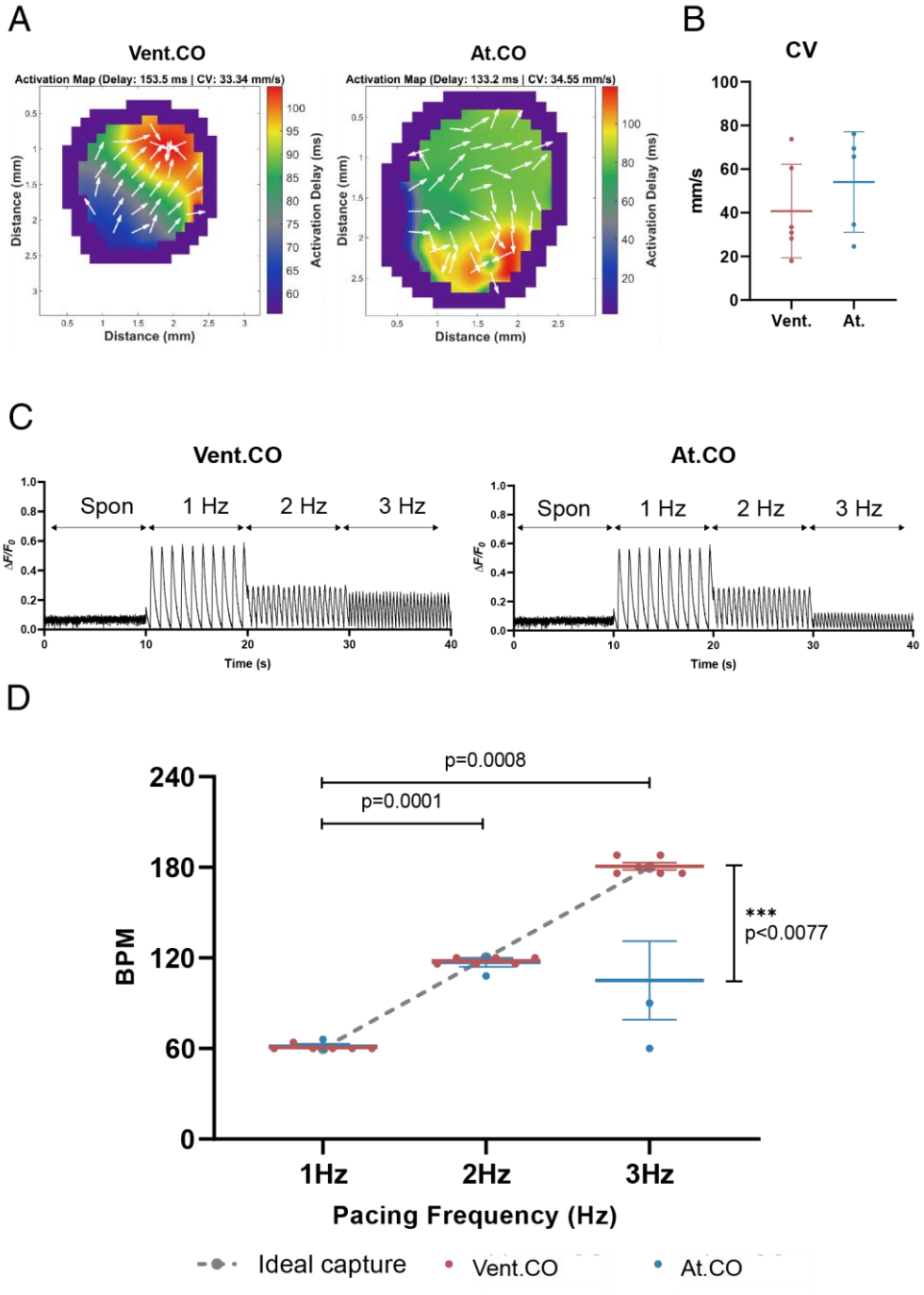
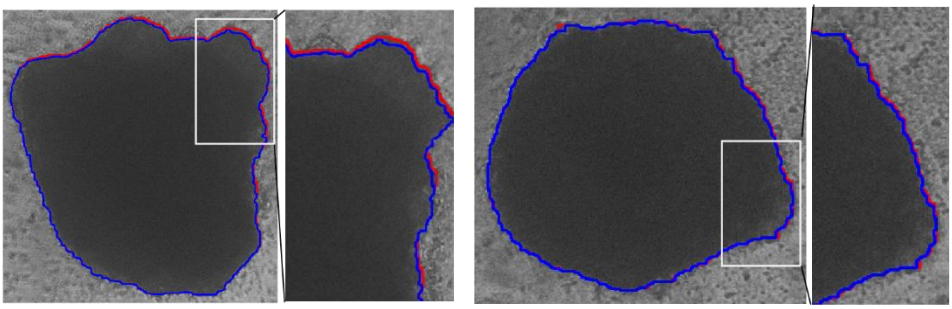

A
Vent.CO
Activation Map (Delay: 153.5 ms | CV: 33.34 mm/s)
Distance (mm)
Distance (mm)
Activation Delay (ms)
At.CO
Activation Map (Delay: 133.2 ms | CV: 34.55 mm/s)
Distance (mm)
Distance (mm)
Activation Delay (ms)
B
CV
mm/s
Vent.
At.
C
Vent.CO
Spon
1 Hz
2 Hz
3 Hz
ΔF/F0
Time (s)
At.CO
Spon
1 Hz
2 Hz
3 Hz
ΔF/F0
Time (s)
D
p=0.0008
p=0.0001
***
p<0.0077
BPM
1Hz
2Hz
3Hz
Pacing Frequency (Hz)
Ideal capture
Vent.CO
At.CO

Fig. 16 Spatiotemporal conduction and pacing fidelity of chamber-specific 3D cardiac organoids. (Top) Representative isochronal activation maps for ventricular and atrial organoids; the colour scale denotes activation delay (ms) and white quiver vectors indicate local conduction direction and magnitude. Macroscopic conduction velocities were comparable between chambers (≈ 33–35 mm/s). (Bottom) Pacing-fidelity analysis under 1–3 Hz field stimulation. Ventricular organoids track the pacing rate along the line of identity through 3 Hz; atrial organoids show greater variability at the highest frequency. vCO, red; aCO, blue.

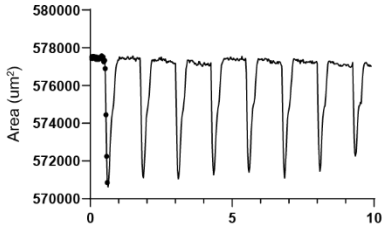
4.8 Label-Free Mechanical Contractility

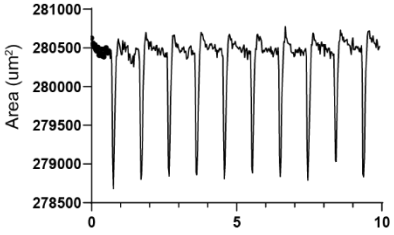
Mechanical function was assessed non-invasively from brightfield recordings using VideoROI3Splitter **(Fig. 17A–B).** Consistent with their respective electrophysiological phenotypes, aCOs beat substantially faster than vCOs (spontaneous rate ≈ 43.9 versus ≈ 15.8 BPM; ***$p < 0.001$), while vCOs exhibited a longer contraction duration (≈ 811 versus ≈ 656 ms) and a larger normalized contraction amplitude **(Fig. 17C–D).** The greater amplitude and longer duration of ventricular contractions, set against the faster cycling of atrial tissue, mirror the Ca^{2+} functional fingerprint and confirm chamber-specific divergence at the mechanical level.

A

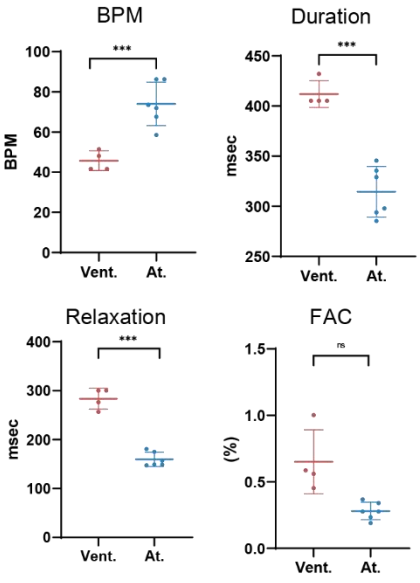


B


Area (um²)
580000
578000
576000
574000
572000
570000
0
5
10


Area (um²)
281000
280500
280000
279500
279000
278500
0
5
10

C


BPM
***
BPM
Vent.
At.
Duration
***
msec
Vent.
At.
Relaxation
***
msec
Vent.
At.
FAC
ns
(%)
Vent.
At.

D

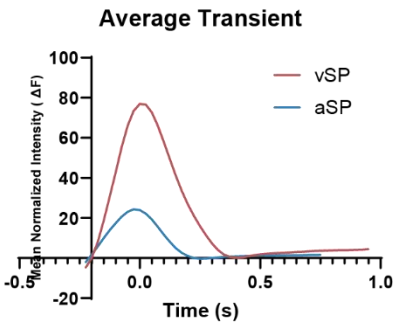

Average Transient
vSP
aSP
Mean Normalized Intensity (ΔF)
Time (s)

Fig. 17 Label-free mechanical contractility of chamber-specific 3D cardiac organoids. (A, B) Automated boundary detection and beating traces for ventricular (vCOs) and atrial (aCOs) organoids extracted by VideoROI3Splitter. (C) Quantification of spontaneous beating rate (BPM), contraction duration, and contraction magnitude. (D) Mean normalized contraction amplitude (ΔF): ventricular organoids generate larger-amplitude contractions than atrial organoids. (E) Representative pacing trace showing capture from spontaneous through 0.5, 1, and 3 Hz. Data are mean ± SEM; ***$p < 0.001$.

4.9 Chamber-Specific Pharmacological Validation

To test physiological fidelity, organoids were challenged with two agents of defined chamber selectivity.

Isoproterenol (ISO), a non-selective β-adrenergic agonist, elicited the expected positive chronotropic, inotropic, and lusitropic responses in both chamber types (**Fig. 18**). Representative Ca^{2+} transients recorded before (–ISO) and after (+ISO) 1 μM ISO showed that **both ventricular and atrial organoids** increased their beating rate and transient amplitude ($\Delta F/F_0$), while calcium-handling kinetics accelerated — time-to-peak (TTP), the decay constant (τ), and CTD90 all shortened (**Fig. 18; n = 5 per chamber, paired t-test**). Because β-adrenergic receptors are expressed in both atrial and ventricular myocardium, this uniform, non-selective responsiveness confirms intact adrenergic signaling and SR Ca^{2+} handling in both lineages and establishes a positive-control benchmark against which the chamber-selective agent can be compared.

Carbachol (CCh), a muscarinic M_2 agonist that activates the acetylcholine-sensitive potassium current ($I_{K,ACh}$), revealed clear chamber selectivity **(Fig. 19).** Because M_2 receptors and $I_{K,ACh}$ are atrial-restricted in the native heart, only aCOs responded: CCh produced a pronounced negative chronotropic and inotropic effect, reducing spontaneous rate and transient amplitude with measurable changes in CTD90, τ, and conduction, in some cases approaching beating arrest at higher doses **(Fig. 19A–D; n = 5, paired t-test).** Ventricular transients remained essentially unaffected. This divergent parasympathetic response recapitulates a defining atrial-specific physiological feature and validates the platform for chamber-selective pharmacology

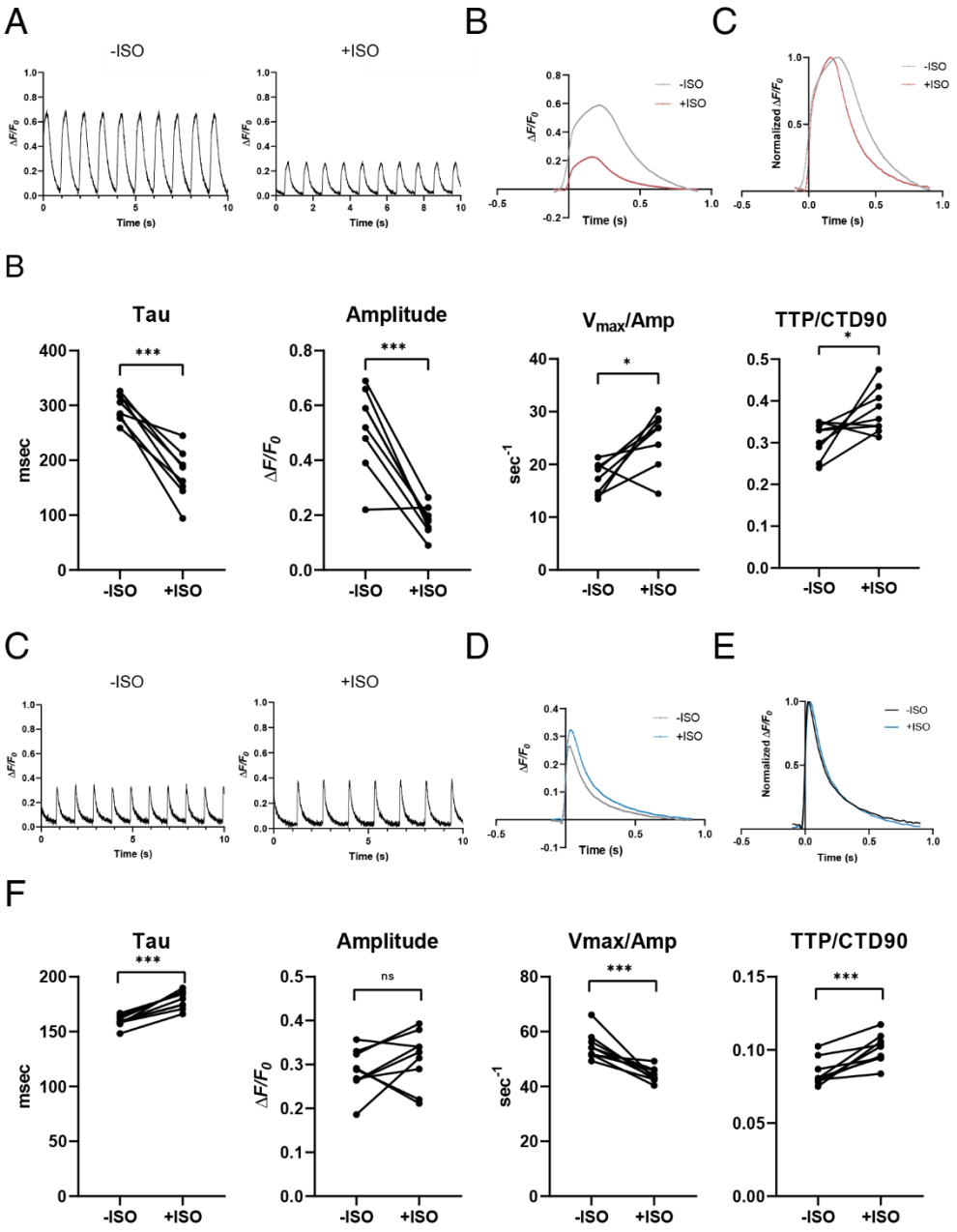

A
-ISO
+ISO
B
C
Time (s)
B
Tau
Amplitude
Vmax/Amp
TTP/CTD90
msec
sec-1
***
*
-ISO
+ISO
C
D
E
F
Vmax/Amp
ns

Fig. 18 Subtype-specific pharmacological sensitivity to Isoproterenol (ISO). Representative Ca^{2+} transients of ventricular (top) and atrial (bottom) organoids before (–ISO) and after (+ISO) treatment with 1 µM ISO. Both chambers show positive chronotropic and inotropic responses with accelerated calcium-handling kinetics (shortened TTP, τ, and CTD90), confirming functional β-adrenergic signaling. Statistical significance by paired t-test (*$p < 0.05$, **$p < 0.01$, ***$p < 0.001$).

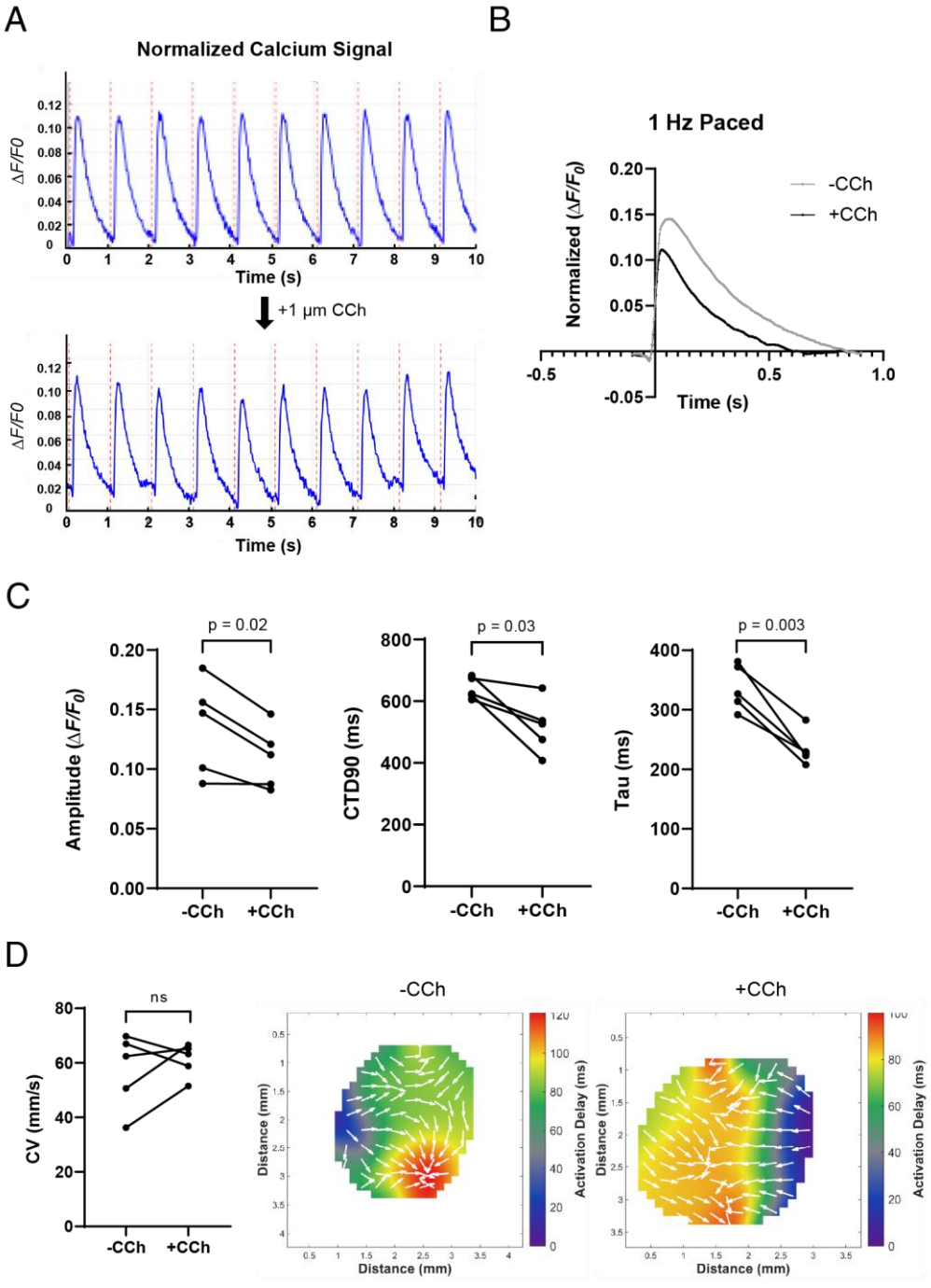

A
Normalized Calcium Signal
ΔF/F0
Time (s)
+1 μm CCh
B
1 Hz Paced
Normalized (ΔF/F0)
-CCh
+CCh
Time (s)
C
p = 0.02
Amplitude (ΔF/F0)
p = 0.03
CTD90 (ms)
p = 0.003
Tau (ms)
-CCh
+CCh
D
ns
CV (mm/s)
-CCh
+CCh
Distance (mm)
Activation Delay (ms)

Fig. 19 Pharmacological validation of atrial-specific functional response to Carbachol. (A) Normalized Ca^{2+} signals of atrial organoids before (–CCh) and after (+1 µM CCh) treatment. (B) Overlaid 1 Hz-paced transients highlighting the negative inotropic effect. (C) Paired quantification of amplitude ($\Delta F/F_0$), CTD90, and τ. (D) Conduction velocity and corresponding activation maps before and after treatment. Carbachol elicited a robust negative chronotropic and inotropic response in atrial organoids, whereas ventricular organoids were unaffected. Data are mean ± SEM (n = 5); paired t-test (*$p < 0.05$, ***$p < 0.001$).

4.10 Proof-of-Concept High-Throughput Multi-Organoid Platform

Finally, compatibility with scaled acquisition was demonstrated on a 3D-printed micro-pillar array housing nine organoids per device **(Fig. 20).** Organoids of reproducible diameter (mean ≈ 1.3 mm) were generated, retained chamber-specific MLC2v/MLC2a identity by 3D confocal imaging, and were stably positioned within the pillar/Matrigel constructs. Within a single field of view, VideoROI3Splitter resolved per-organoid area-tracking traces while iCAMAnalyzer simultaneously generated multi-organoid Ca^{2+} recordings and activation maps. These results establish proof-of-concept for parallel, multi-organoid electro-mechanical recording and support the scalability of the platform toward higher-throughput screening.

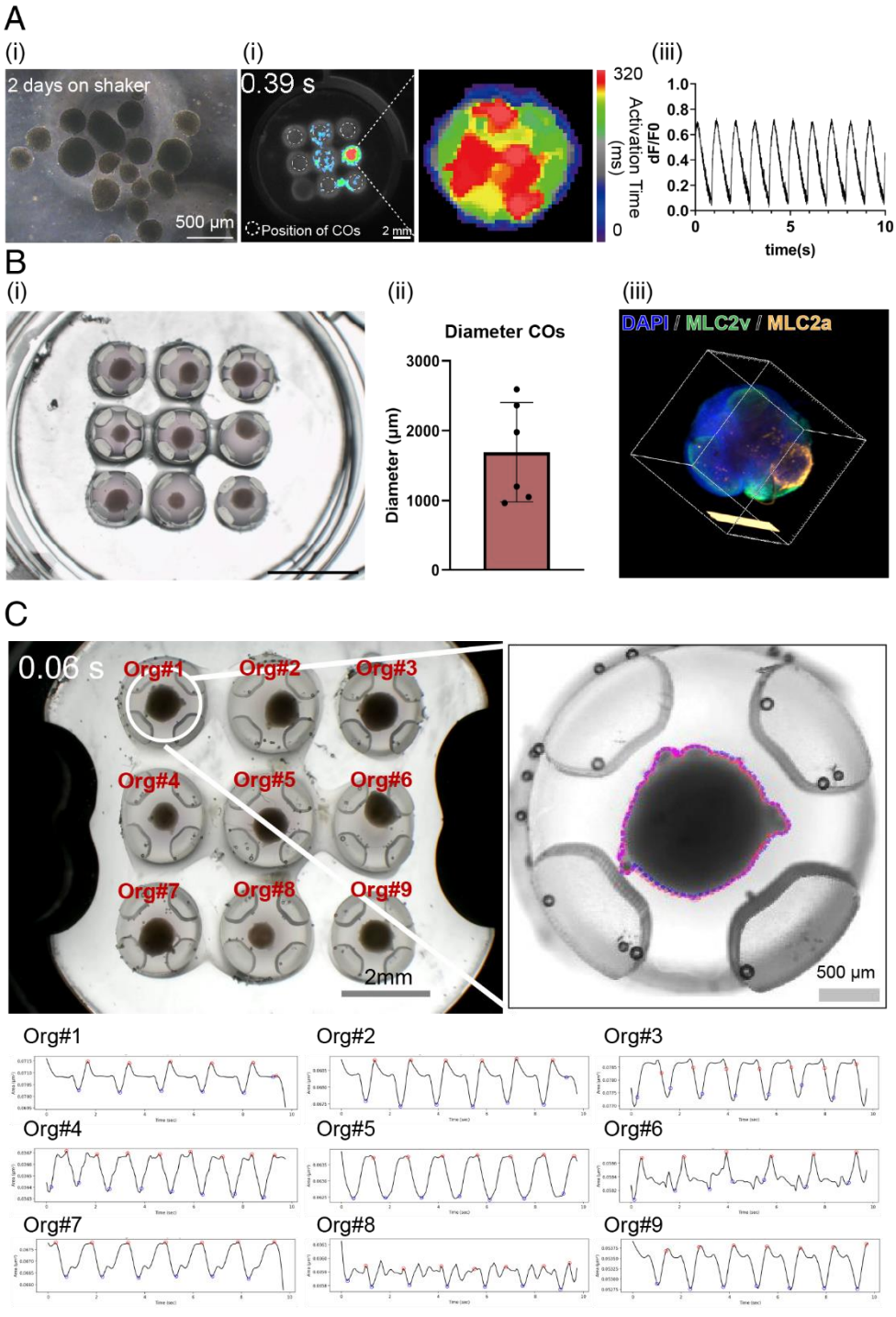

A
(i)
2 days on shaker
500 µm
(i)
0.39 s
Position of COs
2 mm
320
Activation Time (ms)
0
(iii)
1.0
0.8
0.6
0.4
0.2
0.0
dF/F0
0
5
10
time(s)
B
(i)
(ii)
Diameter COs
3000
2000
1000
0
Diameter (µm)
(iii)
DAPI / MLC2v / MLC2a
C
0.06 s
Org#1
Org#2
Org#3
Org#4
Org#5
Org#6
Org#7
Org#8
Org#9
2mm
500 µm
Org#1
Org#2
Org#3
Org#4
Org#5
Org#6
Org#7
Org#8
Org#9

Fig. 20 Proof-of-concept high-throughput, multi-organoid analysis on the micro-pillar platform. A single 3D-printed micro-pillar device houses nine chamber-specific cardiac organoids (COs), enabling parallel structural and functional readout with iCAMAnalyzer.(A) Parallel calcium imaging (Fluorescence mode). *(Left)* COs generated by the shaker/Matrigel method (2 days on shaker; scale bar, 500 μm). *(Middle)* Representative single frame (0.39 s) marking the positions of multiple COs, with the corresponding spatiotemporal activation map (colour scale, activation time in ms; scale bar, 2 mm). *(Right)* Representative $\Delta F/F_0$ calcium-transient trace. (B) Morphology and chamber identity at scale. *(Left)* Brightfield image of nine COs seated in the pillar array (scale bar, 2 mm). *(Middle)* CO diameter (each point = one organoid; bar, mean ± SD). *(Right)* 3D confocal reconstruction of a representative CO immunostained for MLC2v (green) and MLC2a (orange) with DAPI (blue), confirming retention of chamber-specific identity. (C) Parallel label-free contractility (Brightfield mode). *(Left)* Brightfield snapshot (0.06 s) of the nine-organoid field with individually segmented organoids (Org#1–9; scale bar, 2 mm) and a magnified single organoid showing its tracked boundary *(right; scale bar, 500 μm)*. *(Bottom)* Per-organoid area-change (contractility) traces extracted simultaneously for all nine organoids (Org#1–9), demonstrating parallel multi-organoid recording.

V. Discussion

This study established a bioengineering framework for generating hiPSC-derived chamber-specific 3D cardiac organoids (COs) together with a unified, license-free electro-mechanical analysis platform (iCAMAnalyzer). Combining a modified biphasic differentiation protocol with a dual-module Python application, it addresses two persistent bottlenecks: cardiac subtype specification and standardized functional quantification.

5.1 Enhanced Structural and Functional Maturation in 3D Environments

Two-dimensional monolayers cannot reproduce the structural hierarchy and anisotropic alignment of native myocardium, and hiPSC-CMs cultured this way remain structurally immature. In contrast, the 3D COs showed markedly enhanced structural maturation: ventricular organoids developed dense, well-aligned sarcomeres averaging 2.1 μm—approaching the adult ventricular range—with clearly resolved Z-disc registers. This organized, chamber-appropriate architecture demonstrates that transitioning to a three-dimensional culture format substantially improves the structural maturity of hiPSC-derived cardiomyocytes relative to conventional 2D differentiation.

5.2 Preservation of Chamber-Specific Identity via Timely Retinoic Acid Modulation

A common hurdle in 3D cardiac tissue differentiation is the progressive loss of cellular purity and subtype-specific identity over extended culture periods, often yielding heterogeneous cellular pools that display overlapping or highly ambiguous electrophysiological profiles. By implementing a tightly regulated biphasic differentiation regimen—specifically leveraging a restricted temporal window (Days 3–5) for exogenous retinoic acid (RA) administration—we successfully guided the early

mesodermal lineages into segregated posterior Second Heart Field (pSHF) and anterior second heart field/first heart field (aSHF/FHF) progenitor trajectories. The exclusive localized expression of the atrial selector transcript NR2F2 in aCOs and the ventricular transcript IRX4 in vCOs confirms that mimicking the native embryonic morphogen gradient is vital for establishing true chamber-selective in vitro microenvironments.

5.3 Parallel Quantification and Pharmacological Fingerprinting via iCAMAnalyzer

iCAMAnalyzer provided an objective, standardized framework for multi-modal tissue profiling. Implemented entirely in Python, it consolidates fluorescence signal processing, conduction-velocity mapping, and label-free OpenCV motion tracking into a single application. Rather than splitting calcium and contractility analysis across separate, single-purpose programs, both readouts are processed within one unified, license-free Python workflow.

Within this framework, two newly derived parameters—the Shape Index and Efficiency Index—converted qualitative optical traces into reproducible "functional fingerprints," cleanly separating the rapid upstroke and abbreviated decay kinetics of aCOs from the prolonged, plateau-dominated calcium-handling waveforms of vCOs.

The physiological fidelity of these profiles was corroborated pharmacologically: carbachol elicited a robust negative inotropic and chronotropic response exclusively in aCOs, reflecting preserved atrial-specific muscarinic signaling ($I_{K,ACh}$). Such chamber-selective drug actions are routinely missed by generic, non-specific preclinical assays, underscoring the platform's value.

5.4 Limitations and Technical Constraints

Despite the significant structural and analytical advancements achieved in this study, several limitations must be acknowledged:

1. Separate optical acquisition of the two modalities: iCAMAnalyzer analyzes both the calcium (fluorescence) and the mechanical (brightfield motion) readouts within a single, unified Python workflow. The two signals, however, are still acquired as separate high-speed recordings—Fluo-4 fluorescence for calcium and brightfield for motion—rather than from one dual-channel stream. Consequently, true same-beat, real-time excitation–contraction (E–C) coupling latency cannot yet be resolved from a single dataset; this would require simultaneous dual-channel optics (Section 5.5).

2. Lack of Mature T-Tubule Circuitry: While the engineered vCOs demonstrated improved sarcomeric alignment, they still lack a fully integrated, mature transverse-tubular (T-tubule) network. Consequently, calcium release remains partly dependent on peripheral sarcolemmal diffusion rather than the synchronous whole-cell dyadic discharge seen in native adult tissue.

3. Absence of Non-Myocyte Lineages: The current model relies on a highly enriched cardiomyocyte population. The exclusion of non-myocyte stromal lineages—cardiac fibroblasts, endothelial cells, and resident immune cells—limits the platform's ability to model complex multicellular phenotypes such as myocardial fibrosis or inflammatory cardiotoxicity.

5.5 Future Directions

To build upon the foundational platform established in this work, future investigations will focus on the following milestones:

1. **Hardware Integration for Dual-Channel Recording:** Future iterations of the platform will integrate high-speed dual-channel optical splitters onto the imaging path. This hardware modification will allow iCAMAnalyzer to simultaneously capture fluorescence calcium waves and brightfield

contractility frame stacks in real-time from the exact same beat, enabling the definitive mapping of E-C coupling efficiency under varying workloads.

2. **Incorporation of Multicellular Formulations:** To more accurately mimic the cellular landscape of the human heart, future differentiation protocols will incorporate hiPSC-derived cardiac fibroblasts and endothelial cells at physiological ratios during the 3D aggregation phase, enhancing extracellular matrix remodeling and long-term tissue survival.

3. **Scalable Patient-Specific Disease Modeling:** The high concordance and automation supported by the PyQt5 GUI layout make iCAMAnalyzer ideally suited for large-scale screening. We plan to utilize this platform to screen patient-specific iPSC lines derived from clinical cohorts presenting with atrial fibrillation (AF) and ventricular hypertrophic cardiomyopathy (HCM), providing an objective tool for precision medicine and low-cost cardiotoxicity screening.

VI.Conclusion

This study presents a scalable, translationally relevant platform for generating, maturing, and digitally profiling hiPSC-derived chamber-specific 3D cardiac organoids. Its contributions are threefold: (1) a bioengineering pipeline combining a 3D-printed micro-pillar array with a biphasic differentiation protocol to produce high-purity, structurally enhanced atrial (aCO) and ventricular (vCO) organoids; (2) iCAMAnalyzer, a custom, license-free, dual-module Python application unifying calcium-transient kinetics, two-pass conduction-velocity mapping, and multi-ROI motion tracking in one reproducible GUI; and (3) physiological validation through structural (sarcomere alignment), molecular (NR2F2 vs. IRX4), and pharmacological (carbachol-selective atrial response) chamber divergence. By bridging 2D cultures and native cardiac physiology, the platform aligns with the goals of the FDA Modernization Act 2.0 and the CiPA initiative, offering a license-free, scalable tool for precision drug discovery, safety pharmacology, and patient-specific disease modeling.

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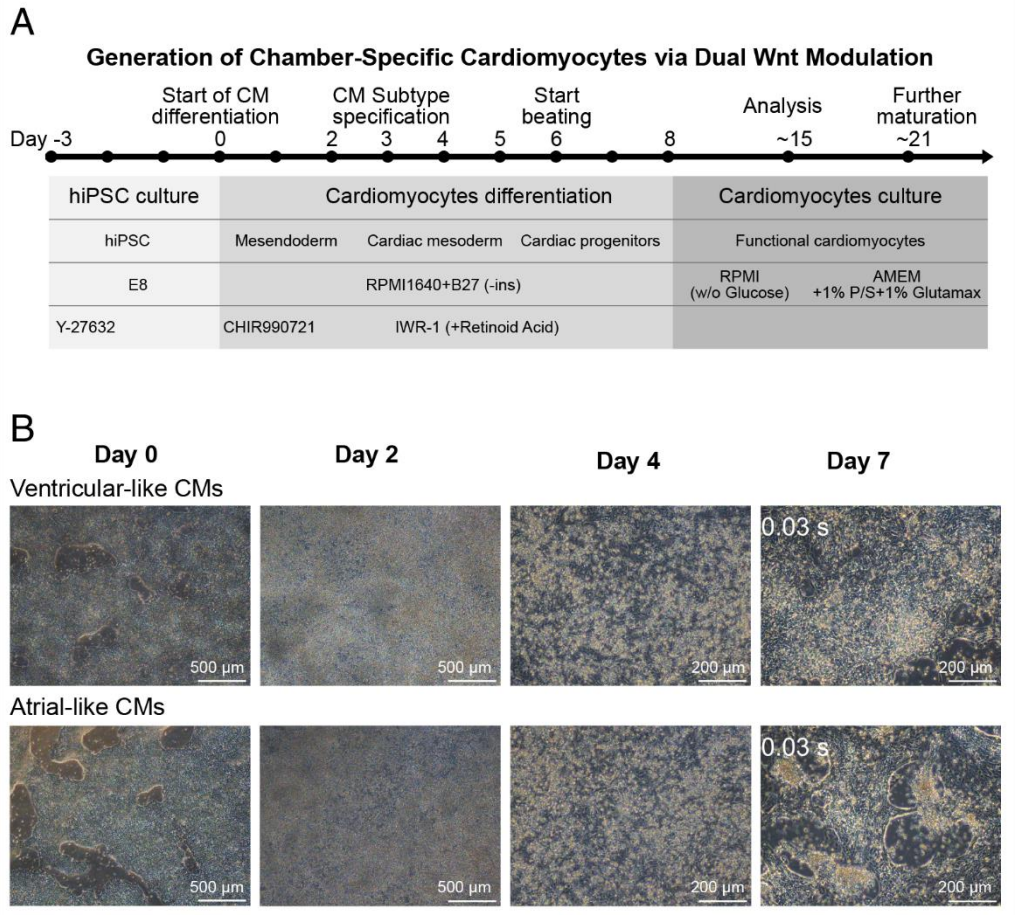
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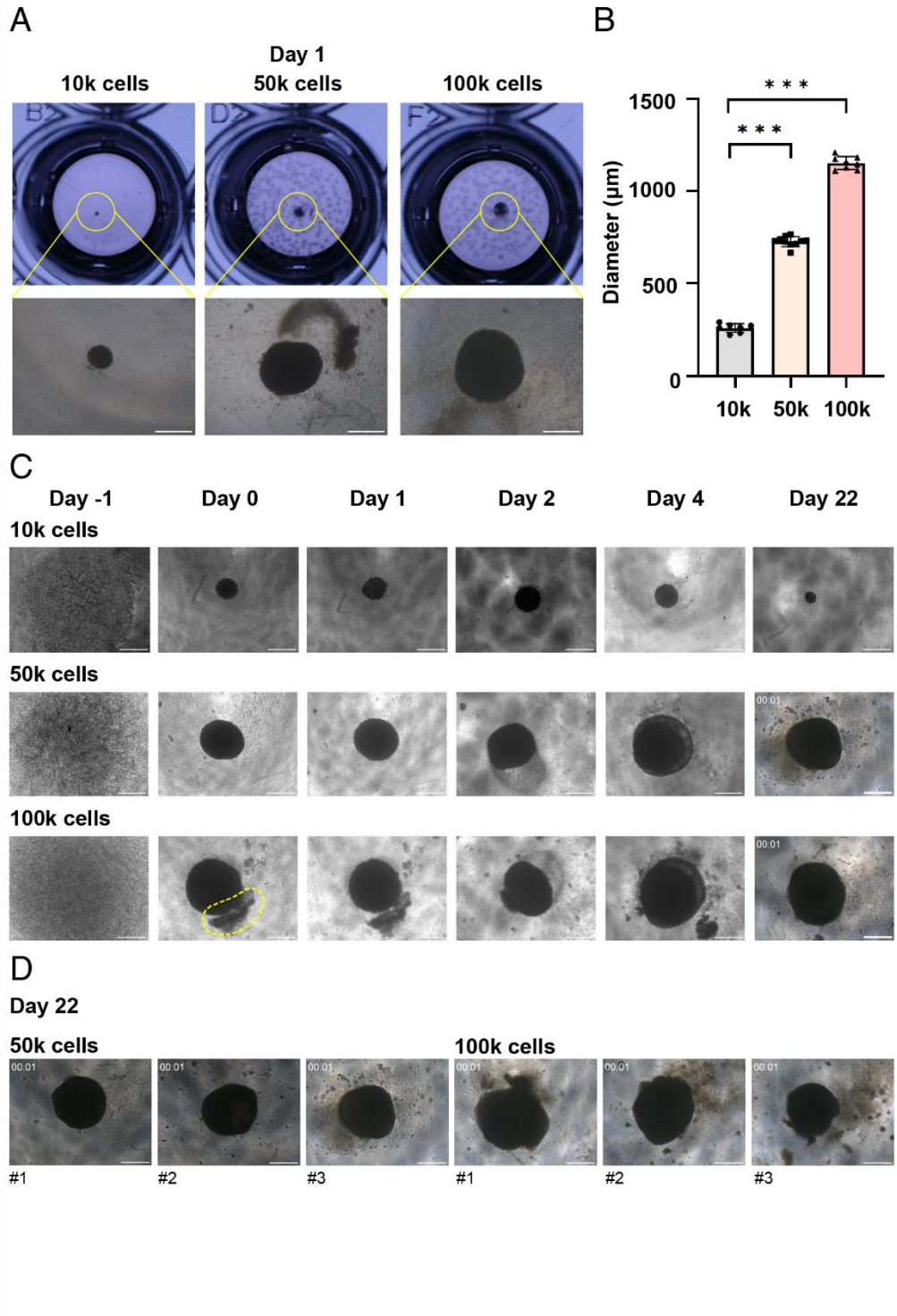
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VIII. Supplementary Figures

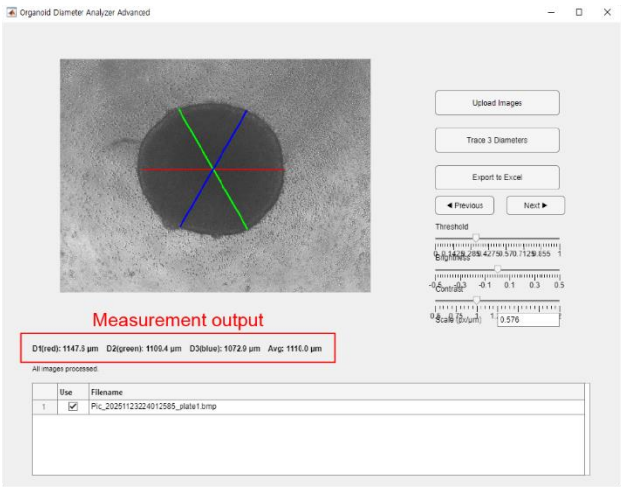


Supplementary Fig. S1 Induction of chamber-specific cardiomyocytes from hiPSCs via dual Wnt signaling modulation. (A) Schematic timeline of the differentiation protocol. Human induced pluripotent stem cells (hiPSCs) were directed toward the cardiac lineage through sequential modulation of the Wnt signaling pathway using CHIR99021 (Wnt activation) and IWR-1 (Wnt inhibition). For atrial-specific specification, Retinoic Acid (RA) was supplemented during the cardiac progenitor stage (Day 4–6). (B) Representative bright-field images showing the morphological transition from hiPSCs to functional cardiomyocytes. By Day 7, both ventricular-like and atrial-like lineages exhibited characteristic elongated and branching cellular morphologies.

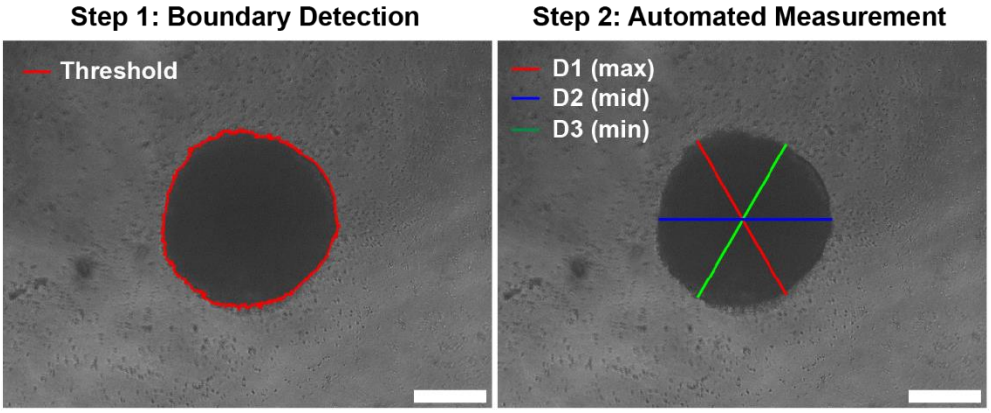

A
Day 1
10k cells
50k cells
100k cells
B
Diameter (µm)
1500
1000
500
0
* * *
* * *
10k
50k
100k
C
Day -1
Day 0
Day 1
Day 2
Day 4
Day 22
10k cells
50k cells
100k cells
D
Day 22
50k cells
100k cells
#1
#2
#3
#1
#2
#3

Supplementary Fig. S2 Optimization of initial cell seeding density for stable cardiac organoid formation. (A) Morphological comparison of cardiac organoids at Day 1 relative to initial seeding densities (10k, 50k, and 100k cells). Insets show magnified views of the self-assembled structures. Scale bars: 500 μm. (B) Quantitative analysis of organoid diameters at Day 1. A significant dose-dependent increase in diameter was observed across all groups (n=6, ***p < .001). (C) Time-course tracking of organoid formation from Day -1 to Day 22. While the 10k and 50k groups achieved stable self-assembly and maintained structural integrity, the 100k group exhibited unincorporated satellite cellular debris early in the process (indicated by the red circle at Day 0). (D) Long-term morphological stability at Day 22. The 100k group demonstrated severe structural heterogeneity and irregular boundaries compared to the uniform spherical morphology of the 50k group. Based on these observations, an intermediate density (e.g., 75k) was empirically selected to balance sufficient contractile mass with long-term structural stability.

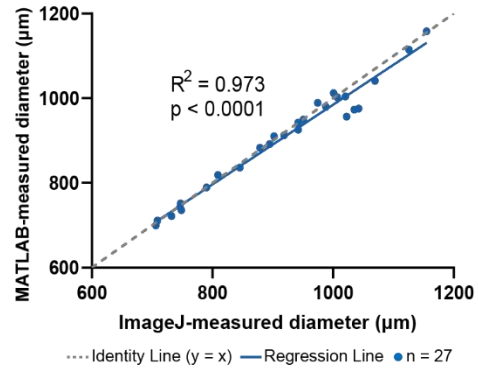
A


Organoid Diameter Analyzer Advanced
Upload Images
Trace 3 Diameters
Export to Excel
◄ Previous
Next ►
Threshold
Brightness
Contrast
Scale (px/µm)
0.576
Measurement output
D1(red): 1147.5 µm D2(green): 1100.4 µm D3(blue): 1072.0 µm Avg: 1106.6 µm
All images processed.
Use
Filename
Pic_20251123224012585_plate1.bmp

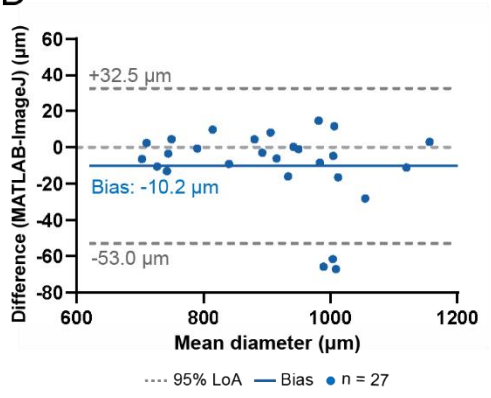
B


Step 1: Boundary Detection
Step 2: Automated Measurement
Threshold
D1 (max)
D2 (mid)
D3 (min)

C

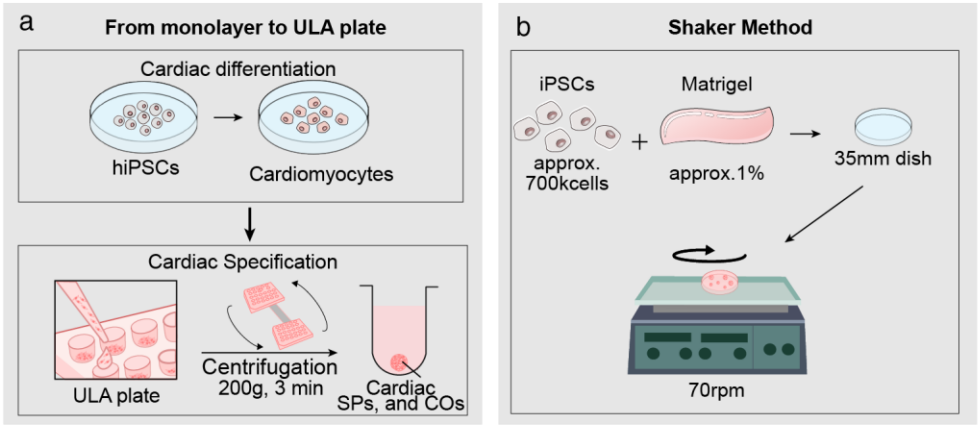

MATLAB-measured diameter (µm)
1200
1000
800
600
R2 = 0.973
p < 0.0001
600
800
1000
1200
ImageJ-measured diameter (µm)
Identity Line (y = x)
Regression Line
n = 27

D

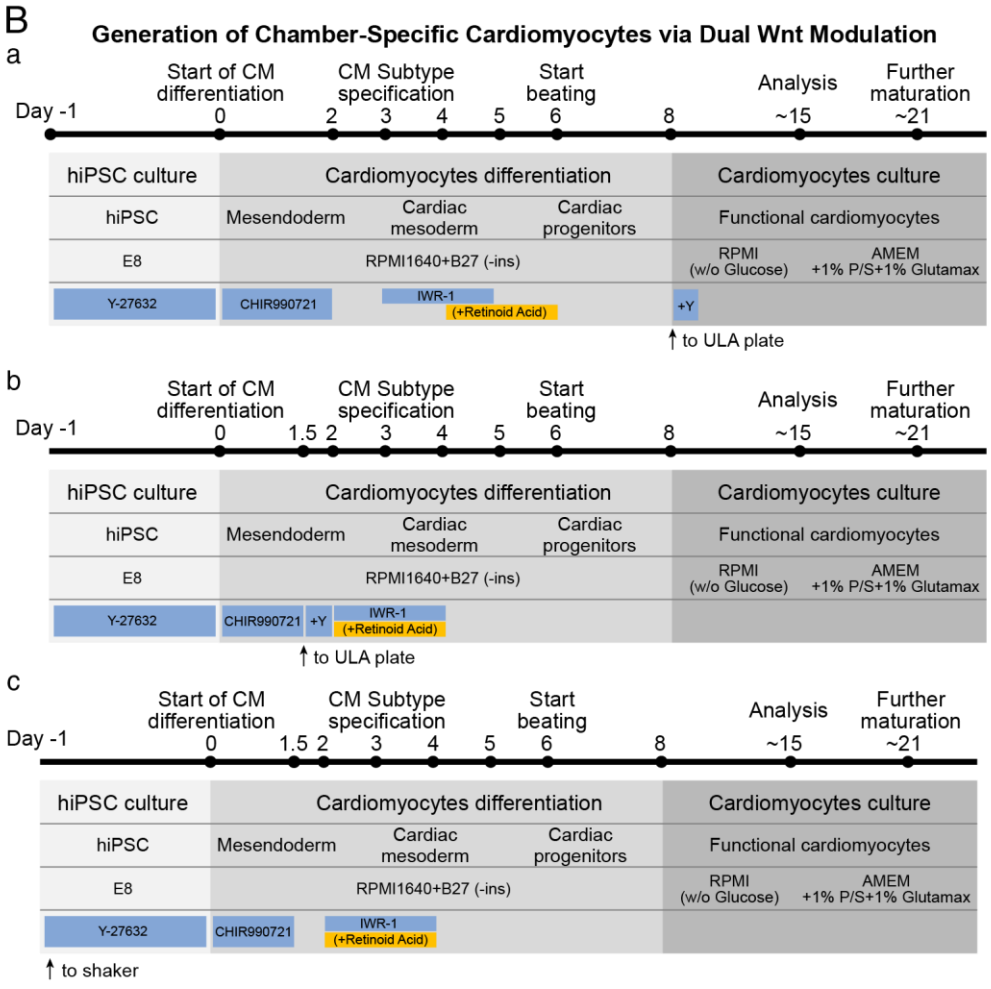

Difference (MATLAB-ImageJ) (µm)
60
40
20
0
-20
-40
-60
-80
+32.5 µm
Bias: -10.2 µm
-53.0 µm
600
800
1000
1200
Mean diameter (µm)
95% LoA
Bias
n = 27

Supplementary Fig. S3 Automated morphological quantification via the custom-developed Organoid Diameter Analyzer. (A) Representative graphical user interface (GUI) of the MATLAB-based "Organoid Diameter Analyzer Advanced." (B) Two-step automated measurement pipeline. Step 1 (Left): Boundary detection isolates the organoid from background using a threshold algorithm. Step 2 (Right): Three diameters are automatically measured — D1 (red, maximum), D2 (blue, intermediate), and D3 (green, minimum) — to derive mean diameter and circularity. (C) Correlation between ImageJ- and MATLAB-derived diameter measurements (n = 27). Dashed line: identity line ($y = x$); solid line: linear regression ($R^2 = 0.973$, $p < 0.0001$). (D) Bland-Altman analysis showing measurement agreement. Bias: −10.2 μm; 95% limits of agreement: −53.0 to +32.5 μm.

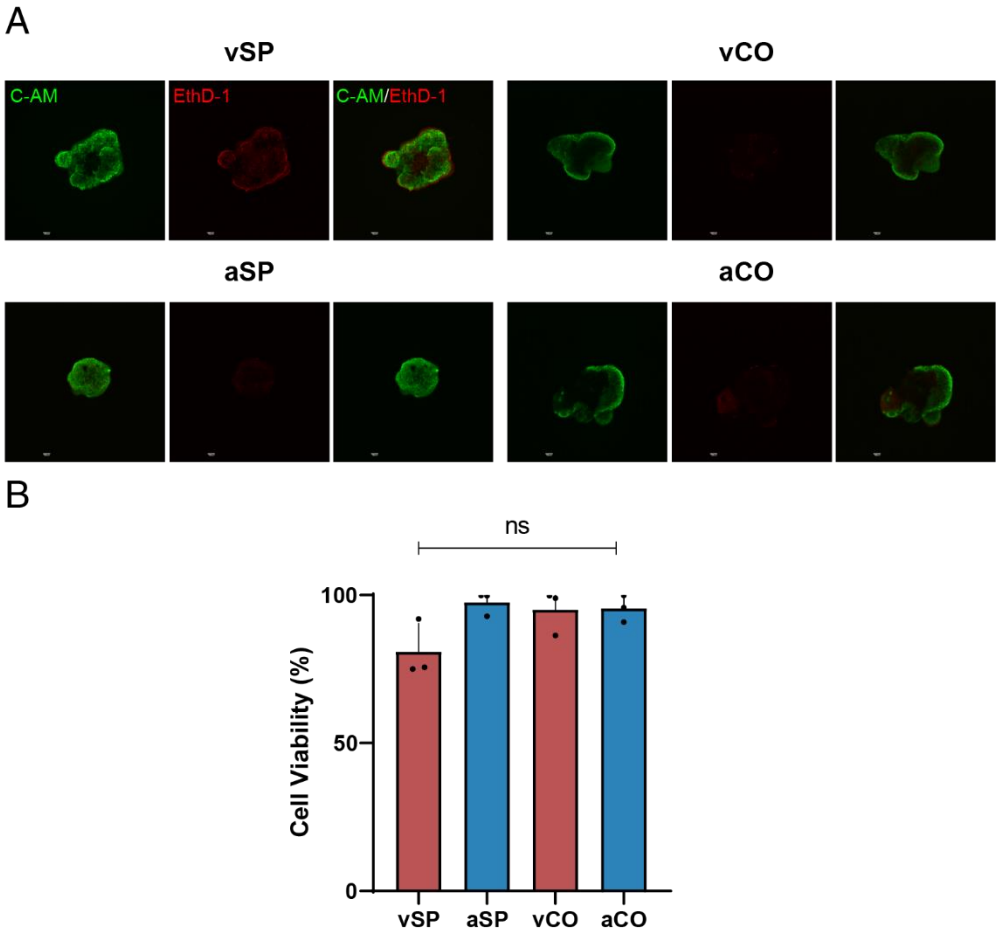
A


a
From monolayer to ULA plate
Cardiac differentiation
hiPSCs
Cardiomyocytes
Cardiac Specification
ULA plate
Centrifugation
200g, 3 min
Cardiac
SPs, and COs
b
Shaker Method
iPSCs
Matrigel
approx.
700kcells
approx.1%
35mm dish
70rpm

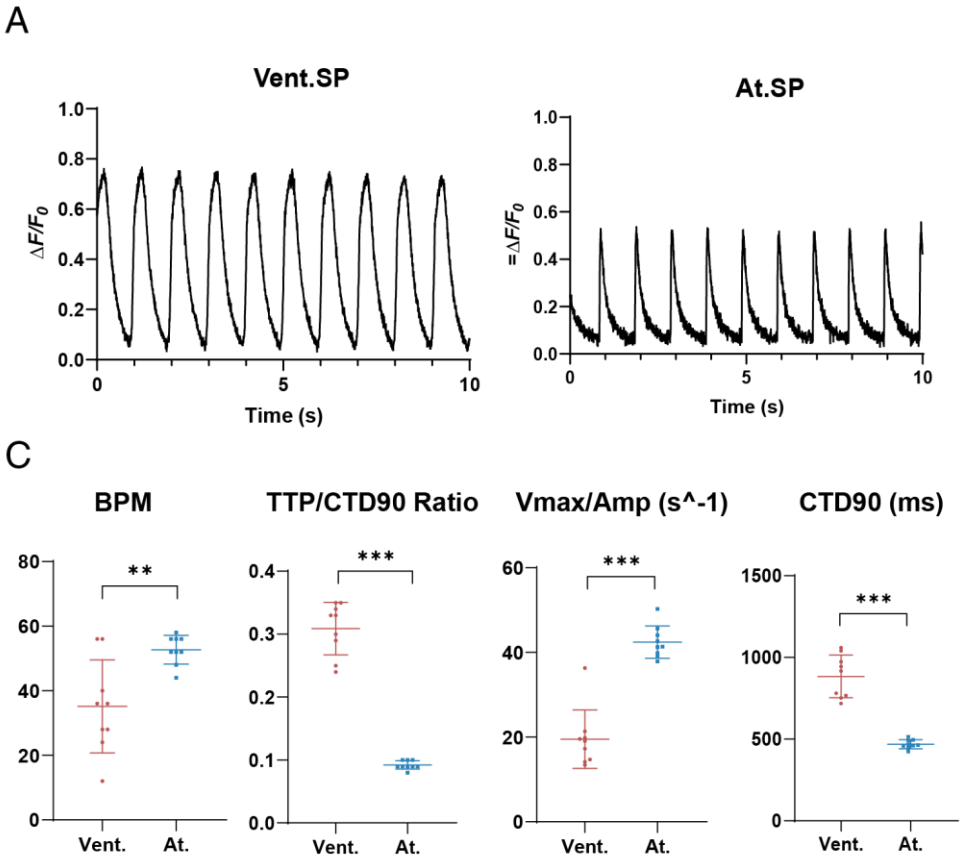
B


Generation of Chamber-Specific Cardiomyocytes via Dual Wnt Modulation
a
Start of CM differentiation
CM Subtype specification
Start beating
Analysis
Further maturation
Day -1
0
2
3
4
5
6
8
~15
~21
hiPSC culture
Cardiomyocytes differentiation
Cardiomyocytes culture
hiPSC
Mesendoderm
Cardiac mesoderm
Cardiac progenitors
Functional cardiomyocytes
E8
RPMI1640+B27 (-ins)
RPMI (w/o Glucose)
AMEM +1% P/S+1% Glutamax
Y-27632
CHIR990721
IWR-1 (+Retinoid Acid)
+Y
to ULA plate
b
Start of CM differentiation
CM Subtype specification
Start beating
Analysis
Further maturation
Day -1
0
1.5
2
3
4
5
6
8
~15
~21
hiPSC culture
Cardiomyocytes differentiation
Cardiomyocytes culture
hiPSC
Mesendoderm
Cardiac mesoderm
Cardiac progenitors
Functional cardiomyocytes
E8
RPMI1640+B27 (-ins)
RPMI (w/o Glucose)
AMEM +1% P/S+1% Glutamax
Y-27632
CHIR990721
+Y
IWR-1 (+Retinoid Acid)
to ULA plate
c
Start of CM differentiation
CM Subtype specification
Start beating
Analysis
Further maturation
Day -1
0
1.5
2
3
4
5
6
8
~15
~21
hiPSC culture
Cardiomyocytes differentiation
Cardiomyocytes culture
hiPSC
Mesendoderm
Cardiac mesoderm
Cardiac progenitors
Functional cardiomyocytes
E8
RPMI1640+B27 (-ins)
RPMI (w/o Glucose)
AMEM +1% P/S+1% Glutamax
Y-27632
CHIR990721
IWR-1 (+Retinoid Acid)
to shaker

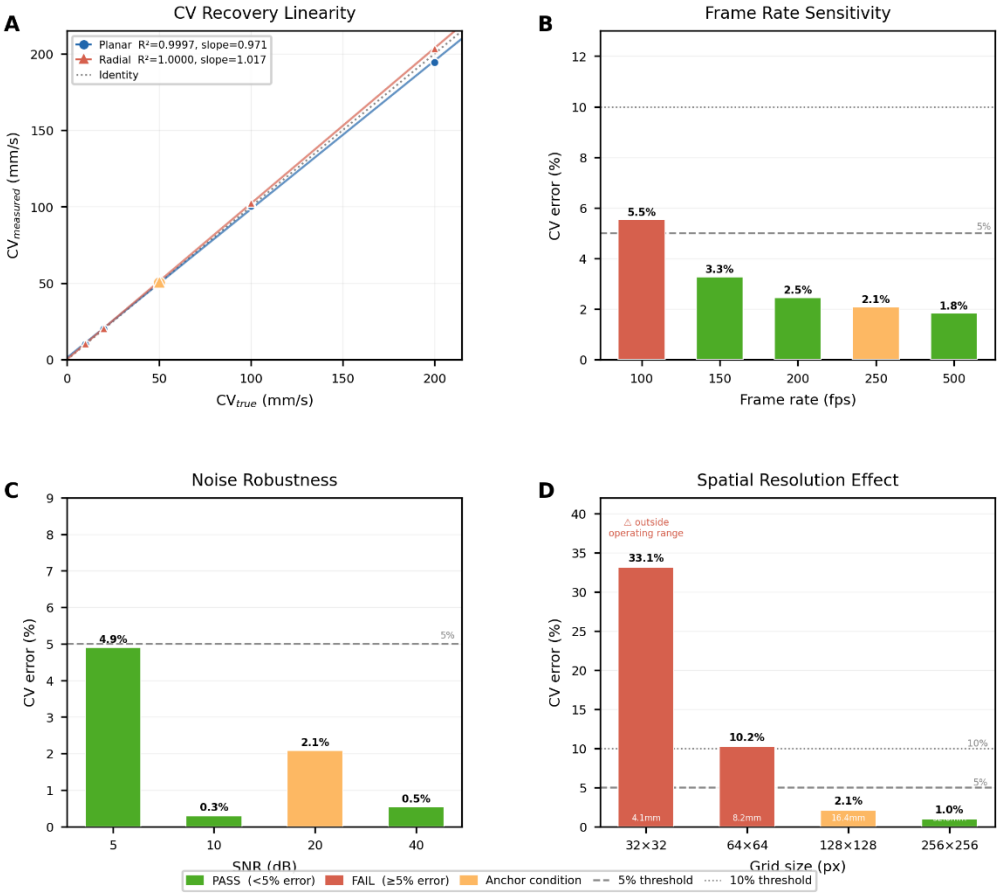
Supplementary Fig. S4 Differentiation timelines and aggregation strategies for chamber-specific cardiac spheroids and organoids. (A) Two aggregation approaches for 3D assembly. (a) Monolayer-to-ULA route: hiPSC-derived cardiomyocytes are dissociated and re-aggregated in ultra-low-attachment (ULA) plates by brief centrifugation (200 g, 3 min) to form cardiac spheroids (SPs) and organoids (COs). (b) Shaker method: hiPSCs (~7×10^5 cells) are mixed with ~1% Matrigel in a 35 mm dish and cultured under orbital agitation (70 rpm). (B) Detailed biphasic protocol timelines (Day −1 to ~Day 21) for the three strategies — (a) monolayer-to-ULA (delayed aggregation), (b) early ULA aggregation, and (c) shaker/Matrigel aggregation — showing culture media and the small-molecule schedule (Y-27632, CHIR99021, IWR-1) with stage-specific retinoic acid for atrial specification.



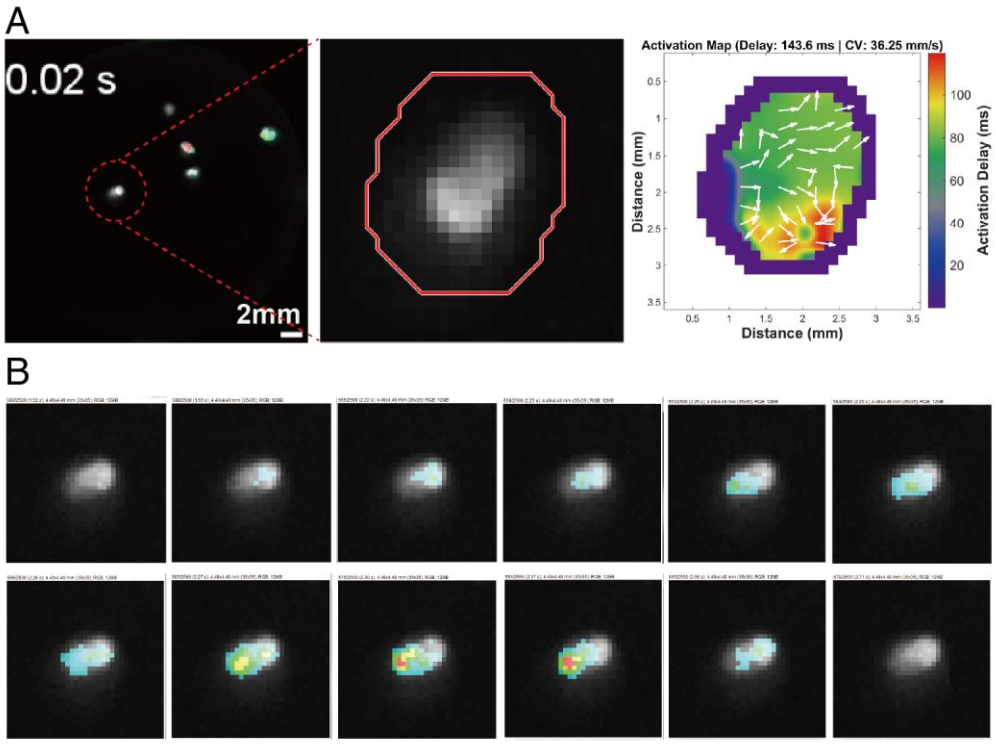
Supplementary Fig. S5 Characterization of cell viability in chamber-specific cardiac organoids. (A) Representative confocal fluorescence images of live-dead assays performed on ventricular Spheroids (vSP), atrial Spheroids (aSP), Microphysiological Ventricular Cardiac Organoids (MvCO), and Microphysiological Atrial Cardiac Organoids (MaCO). Live cells are stained with Calcein-AM (green), and dead cells are identified by Ethidium Homodimer-1 (EthD-1, red). Scale bars: 100 μm (B) Quantitative analysis of cell viability. Viability was calculated as the ratio of live to dead cells using ImageJ. All groups maintained a high viability (mean > 80%), with no statistically significant differences observed between groups (one-way ANOVA, P = 0.059). Data are presented as mean $\pm$ SD (n = 3).



Supplementary Fig. S6 Calcium transient profiling and functional indices of chamber-specific cardiac spheroids. (A) Representative intracellular Ca^{2+} transients ($\Delta F/F_0$) of ventricular (Vent.SP, left) and atrial (At.SP, right) spheroids, showing broad plateau-shaped transients in ventricular versus rapid spike-shaped transients in atrial spheroids. (C) Quantitative comparison between ventricular (Vent.) and atrial (At.) spheroids for beating rate (BPM), Shape Index (TTP/CTD90 ratio), Efficiency Index (Vmax/amplitude, s^{-1}), and 90% calcium-transient duration (CTD90, ms). Atrial spheroids beat faster with a lower Shape Index, higher Efficiency Index, and shorter CTD90, whereas ventricular spheroids show the opposite profile (**p < 0.01, ***p < 0.001).



Supplementary Fig. S7 Quantitative in silico validation of the conduction-velocity and activation-mapping module. (A) Linearity of conduction-velocity (CV) recovery: module-measured versus ground-truth CV for planar (R^2 = 0.9997, slope = 0.971) and radial (R^2 = 1.0000, slope = 1.017) propagation, relative to the identity line. (B) Frame-rate sensitivity: CV error decreases from 5.5% at 100 fps to 1.8% at 500 fps (5% pass threshold indicated). (C) Noise robustness: CV error remains at or below ~5% across signal-to-noise ratios of 5–40 dB. (D) Spatial-resolution effect: CV error decreases with increasing grid size, from 33.1% at 32 × 32 px to 1.0% at 256 × 256 px (5% and 10% thresholds indicated). Colour coding: green, pass (<5% error); red, fail (≥5% error); orange, anchor condition.



Supplementary Fig. S8 Application and cross-method validation of the activation-mapping module on a real cardiac organoid. (A) Left: representative fluorescence field of paced cardiac organoids (t = 0.02 s; scale bar, 2 mm), with one organoid (dashed circle) magnified at centre (red boundary outline). Right: activation map reconstructed by the module (activation delay, 143.6 ms; CV, 36.25 mm/s), with white vectors indicating local propagation direction and the colour scale encoding activation delay (ms). (B) Frame-by-frame time-lapse montage of the Ca^{2+} activation wavefront propagating across the same organoid, independently confirming the propagation direction resolved by the automated module.

국문초록

심방 및 심실 특이적 3D 심장 오가노이드 개발 및 통합 전동-기계 광학 분석 플랫폼 구축

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김영현

심혈관 질환은 전 세계적으로 주요한 사망 원인이지만, 기존 전임상 모델의 낮은 예측 정확도로 인해 효과적인 치료제 개발에 난항을 겪고 있다. 인간 유도만능줄기세포 유래 심근세포(hiPSC-CM)가 인간 중심의 대안을 제공하고 있으나, 표준 2D 배양은 구조적으로 미성숙한 표현형을 보이며, 일반적인 3D 모델은 실제 인간 심장의 복잡한 챔버 특이적 전기생리학적 특성을 재현하는 데 한계가 있다. 또한, 이러한 엔지니어링된 심장 조직의 다각적인 기능적 출력을 객관적으로 정량화할 수 있는 표준화된 통합 분석 플랫폼이 부족한 실정이다. 본 연구에서는 이러한 한계를 극복하기 위해, 특정 분화 단계에서의 레티노산(Retinoic Acid, RA) 조절을 포함하는 Biphasic 분화 프로토콜을 활용하여 hiPSC 유래 챔버 특이적 3D 심장 오가노이드(심방 aCO 및 심실 vCO)를 제작하였다.

제작된 모델을 객관적으로 평가하기 위해, 세포 내 칼슘 동태 및 시공간적 활성 맵핑을 정량화하는 MATLAB 기반의 맞춤형 분석 플랫폼인 *iCAMAnalyzer*를 구축하였으며, 이와 병행하여 라벨프리(Label-free) 수축력 평가를 위한 Python 기반 기계적 운동 추적 모듈을 활용하였다. 구조 및 분자생물학적 평가 결과, 엔지니어링된 vCO 는 약 2.1 μm 의 신장된 근절(Sarcomere) 길이, 뚜렷한 체임버 형태의 빈 공간(Cavity) 형성, 그리고 MLC2v 및 Cx43 의 배타적 발현을 통해 구조적 성숙도를 입증하였다. 반면, aCO 는 약 1.7 μm 의 근절 길이와 함께 MLC2a 및 Cx40 를 발현함을 확인하였다.

*iCAMAnalyzer*를 사용하여, 본 연구는 Shape 및 Efficiency 지표(Indices)를 통해 정량적인 "기능적 지문(Functional fingerprints)"을 확립하였다. vCO 는 안정적인 고원기(Plateau) 형태의 칼슘 파형(높은 Shape 지표)을 보인 반면, aCO 는 빠르고 날카로운 스파이크(Spike) 형태의 활성화 동태(낮은 Shape 지표, 높은 Efficiency 지표)를 나타냈다. 이러한 모델들의 생리학적 타당성은 챔버 특이적 약물 반응을 통해 추가로 검증되었으며, 특히 부교감신경 작용제인 카바콜(Carbachol)은 aCO 에서 선택적인 음성 변력(Inotropic) 및 변시(Chronotropic) 반응을 유도하였다. 마지막으로, 3D 프린팅 기반 마이크로 필라(Micro-pillar) 플랫폼과의 개념 검증(Proof-of-concept) 호환성을 통해 대용량 다중 오가노이드 동시 기록을 위한 시스템의 확장성을 증명하였다. 결론적으로, 본 통합 전기-역학 광학 분석 플랫폼은 정밀 심독성 평가 및 환자 맞춤형 질환 모델링을 위한 객관적이고 생리학적으로 타당한 전임상 프레임워크를 제공한다

주요어: 체외 심장 모델, 심혈관 질환, 조직 공학, 유도만능줄기세포, 질환 모델링, 약물 테스트

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마지막으로, 제가 어떤 길을 걷든 언제나 저를 믿어주고 무한한 사랑과 지지로 든든한 버팀목이 되어준 사랑하는 가족들에게 감사의 마음을 전합니다. 가족들의 따뜻한 지지가 있었기에 지치지 않고 앞으로 나아갈 수 있었습니다.

지금의 결과는 결코 저 혼자만의 힘으로 이룬 것이 아님을 가슴 깊이 새기며,
앞으로 더욱 정진하고 발전하는 모습으로 모든 분들의 은혜에 보답하겠습니다.
다시 한번 저와 함께해 주신 모든 분들께 진심으로 감사드립니다.